



Visvesvaraya Technological University

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A PROJECT PHASE-2 REPORT

ON

**“CHAOTIC MAP BASED IMAGE ENCRYPTION WITH SCAN
PATTERN TECHNIQUE ”**

Submitted to Visvesvaraya Technological University in partial fulfillment of the requirement for the award of Bachelor of Engineering degree in Computer Science and Engineering.

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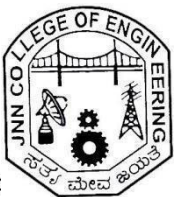
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December 2025

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CERTIFICATE

This is to certify that the project entitled

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ABSTRACT

With the increasing reliance on digital communication and storage of sensitive visual information, securing image data has become paramount. This work proposes an approach to image encryption by integrating chaotic maps and a dynamic scan pattern. The utilization of chaotic maps, known for their pseudorandom and unpredictable behavior, contributes to the robustness of the encryption scheme. The chaotic map is employed to generate a set of cryptographic keys, introducing an additional layer of complexity to the encryption process. These keys dynamically influence the encryption algorithm, enhancing the security of the image data. Furthermore, a unique scan pattern is introduced to shuffle the pixel values during encryption. The proposed encryption method begins by employing a chaotic map to dynamically generate a set of cryptographic keys, adding an element of unpredictability to the encryption process. Subsequently, a unique scan pattern is applied to the image pixels, redistributing them in a manner that enhances the diffusion of information.

ACKNOWLEDGEMENT

On presenting the Project report on “**Chaotic Map Based Image Encryption with Scan Pattern Technique**”, we feel great to express our humble feeling of thanks to all those who have helped us directly or indirectly in the successful completion of the project work.

We would like to thank our respected guide **Mr. Sandeep B**, Assistant Professor, Dept of CS&E, for his continuous encouragement and guidance. We would like to thank our project coordinators, **Dr. Benkappa S M**, **Mrs. Rashmi V**, Department of CSE for all their support and encouragement.

We would like to thank **Dr. Jalesh Kumar**, Professor and Head of Dept of CS&E, JNNCE, Shivamogga, **Dr. Y Vijaya Kumar**, Principal, JNNCE, Shivamogga for their support and encouragement.

We are grateful to Department of Computer Science and Engineering and our institution JNN College of Engineering for imparting us the knowledge with which we could do our best.

Finally, we would like to thank the whole teaching and non-teaching staff of Computer Science and Engineering Department.

Thanking You

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CHAPTER 1

INTRODUCTION

With the fast development of innovations in data communication, it can end up crucial for private information security from prohibitive actions or attackers. Data exchange is closely related to existence, such as instruction, commerce, financial matters, military, e-learning, phone keeping money, and news telecasting. With the modern telecommunication and multimedia technologies progression, a huge amount of critical information voyages in a daily monotony through the shared and open networks. To keep security, sensitive and critical information ought to be secured before conveyance. For data transmitting through any uncertain channel, certain cryptographic techniques are required to change over the coherent information to incomprehensible form before transmitting (encryption). The modern strategies of cryptography are effective for text and image information. However, due to the high redundancy and bulk information capacity, they fail to provide computational security.

The exponential growth in digital communication and storage of images has accentuated the need for reliable and secure encryption techniques to protect sensitive visual information from unauthorized access and tampering. As images play a pivotal role in numerous applications, ranging from medical imaging to secure communication, ensuring the confidentiality and integrity of visual data has become paramount. In response to the ever-evolving landscape of cyber threats, this work introduces an approach to image encryption that harnesses the synergies between chaotic map dynamics and a novel scan pattern [1].

The traditional encryption algorithm such as Data Encryption Standard (DES), Advanced Encryption Standard (AES), and Rivest, Shamir, & Adleman (RSA) are appropriate for text encryption but not appropriate for image encryption for two causes. The first cause is the size of the image is much bigger than the size of the text, so the traditional cryptosystems take more processing time to encrypt the image. The second cause is that the size of decrypted and original image must be equivalent.

Chaos-based encryption is one of the foremost important security technologies within the advanced encryption zone [9]. Chaos hypothesis is created by mathematicians and physicists. Chaos hypothesis has qualified features as nonlinearity, deterministically, abnormality, and

affectability to beginning conditions. Security investigative community receives chaos hypothesis in modern cryptography. To apply a chaos map, there are two ways in a cipher system:

- (i) Produce pseudorandom stream utilizing chaotic maps.
- (ii) Utilize the plain or secret key(s) as control parameters and the introductory conditions.

Finally, apply a few emphases on chaotic systems to get cipher data. The first way compares to stream cipher and the second to block ciphers. The implementation of chaotic maps within the improvement of cryptography systems lies within the truth that a chaotic outline is characterized by

- (i) The beginning conditions and control parameters with high sensitivity.
- (ii) Unpredictability of the orbital advancement.
- (iii) The straight forwardness of the hardware and software execution that leads to a high encryption rate.

1.1 CHAOTIC MAP

Chaotic maps are mathematical models that exhibit chaotic behavior. Chaotic systems are characterized by extreme sensitivity to initial conditions, which means that small changes in the starting conditions can lead to dramatically different outcomes. In mathematics, a chaotic map is a map (namely, an evolution function) that exhibits some sort of chaotic behavior. Maps may be parameterized by a discrete-time or a continuous-time parameter. Discrete maps usually take the form of iterated functions. Chaotic maps often occur in the study of dynamical systems. A chaotic map refers to a mathematical function or system that exhibits chaotic behavior, characterized by sensitivity to initial conditions, irregular and unpredictable dynamics, and a high degree of complexity. These maps are commonly used in the field of dynamical systems theory to model various phenomena, such as population dynamics, weather patterns, and cryptographic algorithms. The hallmark of chaotic maps is their deterministic nature, meaning that their future states are entirely determined by their current state and the governing equations. However, even small variations in the initial conditions can lead to vastly different outcomes over time, making long-term predictions challenging. Chaotic maps have found applications in diverse fields, including physics, biology, and computer science, due to their ability to generate pseudorandom sequences and simulate

complex, unpredictable behavior. Studying chaotic maps provides insights into the fundamental principles of chaos theory, revealing the intricate and often counterintuitive dynamics underlying seemingly random systems. Arnold cat map operates on a square grid of points, typically represented by a 2D array of integers. The transformation involves a series of mathematical operations (matrix multiplication) that reshuffle the positions of points on the grid. The resulting pattern exhibits chaotic behavior, meaning that small changes in the initial conditions can lead to significantly different outcomes over time. Arnold's cat map is shown in the Fig 1.1.

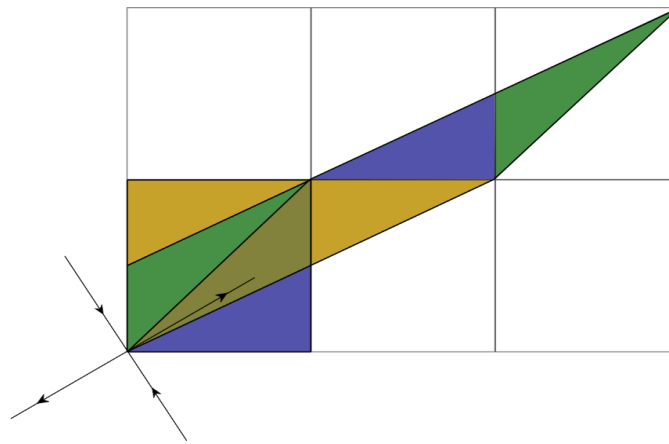


Fig 1.1: Arnold cat map

The logistic map is a mathematical function that is often used to model population growth or the behavior of certain dynamic systems as shown in Fig 1.2. It is a simple recurrence relation that exhibits complex and chaotic behavior under certain conditions.

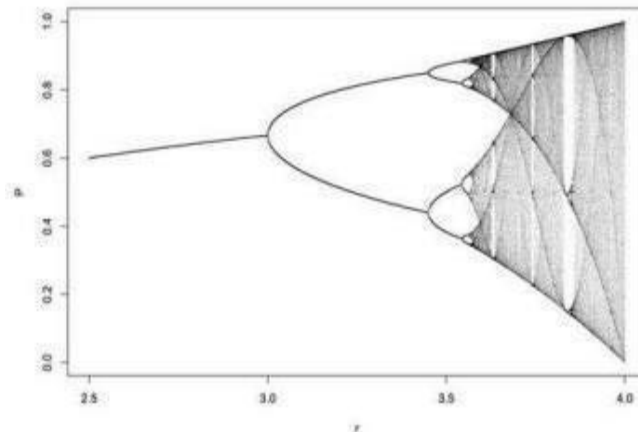


Fig 1.2: Bifurcation Logistic map

The techniques focused on chaos are considered effective in managing with voluminous, excess information. They give quick, profoundly secure strategies of encryption. Due to the important properties of chaos signals, they are considered good for image encryption and increases the robustness of cryptosystem against statistical attacks [2]. These properties include pseudo-random (Chaotic motion is predictable in very short term but appears random for longer periods), non-periodicity (A chaotic system may have sequence of variable values for evolving variables that exactly repeat themselves, such periodic sequence is repelling rather than attracting. So, the variable outside the sequence will diverge from it) and highly sensitive to system parameters and initial conditions (A small changes or differences in initial conditions are persistently magnified because of the dynamics of system), etc. Therefore, combining cryptography with Scan Pattern is considered one of the important fields in information security [1]. The chaos-based encryption algorithm led to new efficient ways to develop images security schemes to satisfy the demand secure image transmission over the communication channel.

Chaotic maps are mathematical models that exhibit chaotic behavior. Chaotic systems are characterized by extreme sensitivity to initial conditions, which means that small changes in the starting conditions can lead to dramatically different outcomes. In mathematics, a chaotic map is a map (namely, an evolution function) that exhibits some sort of chaotic behavior. Maps may be parameterized by a discrete-time or a continuous-time parameter. Discrete maps usually take the form of iterated functions. Chaotic maps often occur in the study of dynamical systems. A chaotic map refers to a mathematical function or system that exhibits chaotic behavior, characterized by sensitivity to initial conditions, irregular and unpredictable dynamics, and a high degree of complexity. These maps are commonly used in the field of dynamical systems theory to model various phenomena, such as population dynamics, weather patterns, and cryptographic algorithms. The hallmark of chaotic maps is their deterministic nature, meaning that their future states are entirely determined by their current state and the governing equations. However, even small variations in the initial conditions can lead to vastly different outcomes over time, making long-term predictions challenging. Chaotic maps have found applications in diverse fields, including physics, biology, and computer science, due to their ability to generate pseudorandom sequences and simulate complex, unpredictable behaviour. Studying chaotic maps provides insights into

the fundamental principles of chaos theory, revealing the intricate and often counterintuitive dynamics underlying seemingly random systems [9].

A "scan pattern" refers to the systematic and organized approach employed by devices or systems when examining a specific area or dataset. This concept finds application in diverse fields such as imaging, radar technology, computer vision, data storage, and search algorithms. In imaging and computer vision, scan patterns dictate how an imaging device captures or analyses an image, with raster scans moving across the image in a grid-like fashion [1]. Radar systems utilize various scan patterns, like conical scans or phased array scans, to direct radar beams effectively. Data storage systems may implement different scan patterns for optimal data retrieval or recording on storage media. In computer science, scan patterns are integral to search algorithms, influencing how data is explored and located. Moreover, in the realm of human vision, scan patterns describe the orchestrated movements of the eyes when observing a scene, and these patterns are analyzed in eye-tracking studies to comprehend visual information processing.

1.2 SCAN PATTERN

The scan method is a formal language-based, two-dimensional spatial accessing methodology capable of representing and generating a wide variety of scanning paths. The SCAN language utilizes four fundamental scan patterns: continuous raster (C), continuous diagonal (D), continuous orthogonal (O), spiral (S), Right orthogonal (A), diagonal parallel (E), vertical symmetry (S), continuous orthogonal (L), zeta (Z), block band raster (R) [1].

A spiral scan pattern refers to a method of systematically covering an area or space in a spiral trajectory as shown in the Fig 1.3. This pattern is often employed in various applications, including radar scanning, imaging systems, and robotic navigation.

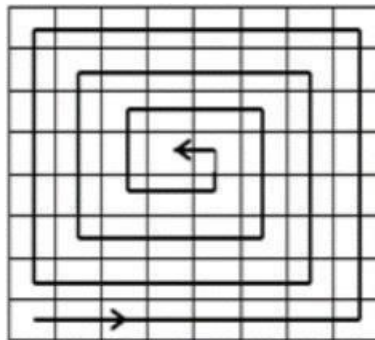


Fig 1.3: Spiral In scan pattern

Zeta-Z scan is a technique used to study the nonlinear optical properties of a material. This method involves scanning a sample along the optical axis (Z-axis) of a focused laser beam as shown in the Fig 1.4.

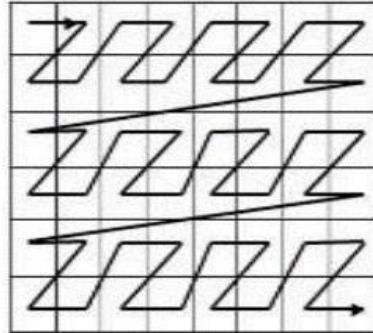


Fig 1.4: Zeta Z scan pattern

Block-B scan pattern refers to something specific, such as a scanning pattern in radar systems, computer programming, or another domain, please provide more context or specify the field. This will help me better understand and provide you with relevant information the block -b is shown in the Fig 1.5.

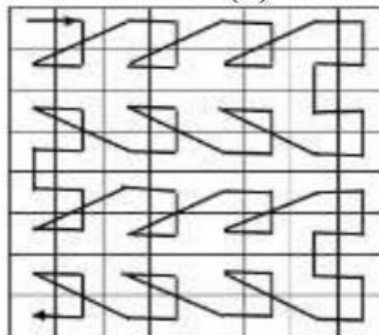


Fig 1.5: Block B scan pattern

Raster scanning is a method used in various fields, including imaging, lidar, and scanning electron microscopy. It involves moving a sensor or beam systematically across an area in a grid-like pattern as shown in the Fig 1.6, typically in rows and columns.

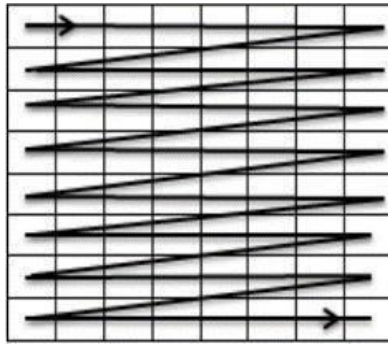


Fig 1.6: Raster R scan pattern

Right Orthogonal -A scan patterns often involve scanning an area in a systematic and organized manner, typically moving along rows and columns or in a grid-like fashion as shown in the Fig 1.7.

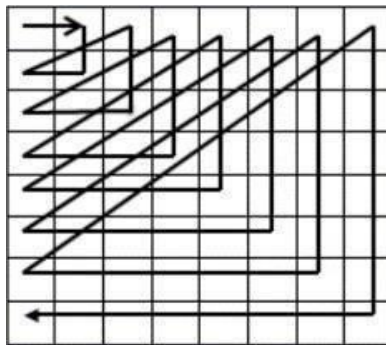


Fig 1.7: Right orthogonal scan pattern

A diagonal-D scan pattern could imply a scanning approach where the scanning device moves diagonally across the target area in the shape of the letter D as shown in the Fig 1.8.

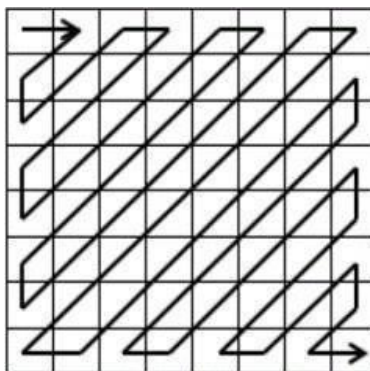


Fig 1.8: Right orthogonal scan pattern

A continuous raster-scan pattern is a method used in imaging and display technologies as shown in Fig 1.9, particularly in the context of cathode-ray tube (CRT) displays and laser scanning systems. The term raster refers to the grid of pixels or points that make up the image. The continuous raster-scan pattern ensures a smooth and orderly coverage of the entire display or imaging area, resulting in a coherent and visually consistent image.

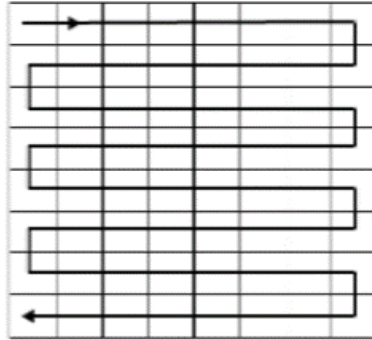


Fig 1.9: Continuous raster scan pattern

The Diagonal Parallel-E scan pattern is a specific type of scan pattern used in parallel computing for traversing a two-dimensional array or matrix as shown in the Fig 1.10. It is essentially a variation of the Parallel-E scan pattern, but instead of moving horizontally across rows, it moves diagonally.

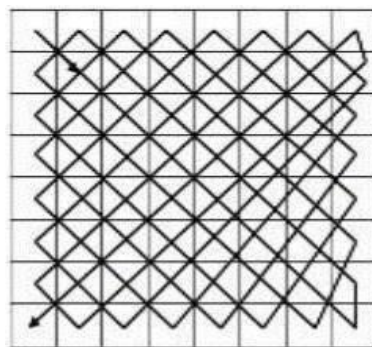


Fig 1.10: Diagonal Parallel-E scan pattern

However, in image encryption algorithms the confusion and diffusion processes Applied successfully [6]. Confusion means that each binary digit (bit) of the cipher image should depend on several parts of the key, obscuring the connections between the two. The property of confusion hides the relationship between the cipher image and the key. This property makes it difficult to find the key from the cipher image and if a single bit in a key is changed, the calculation of most or all the bits in the cipher image will be affected. Confusion increases the ambiguity of cipher image and it is used by both block and stream ciphers.

Diffusion means that if a single bit of the plain image is changed, then about half of the bits in the cipher image should change, and similarly, if one bit of the cipher image is changed, then about half of the plain image bits should change. This is equivalent to the expectation that encryption schemes exhibit an avalanche effect[6]. The purpose of diffusion is to hide the statistical relationship between the cipher image and plain image. To better encrypt the image, the encryption scheme performs two rounds of confusion and diffusion operations. From simulation results, it can be analyzed that the developed method can effectively encrypt different types of images and has a strong ability to withstand a variety of security attacks and data loss attacks.

1.3 PROBLEM DESCRIPTION

In the modern digital landscape, the security of digital images is a matter of great concern due to the ever-increasing volume of sensitive visual data being generated and shared. Protecting these images from unauthorized access and tampering requires innovative and robust techniques.

The traditional encryption algorithm such as Data Encryption Standard (DES), Advanced Encryption Standard (AES), and Rivest, Shamir, & Adleman (RSA) are appropriate for text encryption but not appropriate for image encryption for two causes. The first cause is the size of the image is much bigger than the size of the text, so the traditional cryptosystems take more processing time to encrypt the image. The second cause is that the size of decrypted and original image must be equivalent.

Here are some reasons why one might choose scan pattern and chaotic map-based approaches over AES and DES for image security, Scan pattern and chaotic maps can be designed to influence the visual appearance of the encrypted image. This is particularly

beneficial when the encrypted data is an image, as maintaining visual confidentiality is often crucial. AES and DES, being block ciphers, may not preserve the visual characteristics of the image during encryption. The behavior of a chaotic system is unpredictable. Therefore, it resembles noise. Hence, it is required to design a secure chaotic based encryption system which takes images as input and encrypts the image into cipher image.

1.4 OBJECTIVES

Following are the Objectives of the project undertaken:

- To develop an encryption system using scan patterns and chaotic maps.
- Implement Zeta-Z scan to rearrange pixel positions.
- Use Arnold Cat Map to introduce chaotic confusion.
- Integrate all stages into a single encryption–decryption system.
- Perform evaluation of cryptosystem.

1.5 ORGANIZATION OF THE REPORT

The further report includes following contents, Chapter 2 consists of literature survey of different reference papers, Chapter 3 represents system design and implementation of the proposed work, Chapter 4 consists of result and analysis of the proposed work and Chapter 5 consists of conclusion and future scope of the proposed work.

CHAPTER 2

LITERATURE SURVEY

In this chapter, survey of various journals and papers on chaos-based image encryption algorithms are explained along with brief description, working of the algorithm, system design, advantages, and disadvantages.

2.1 “Image Encryption using Scan Patterns and Hill Cipher”

Authors: V Praveena , P Hari Manohar Varma and BVSSN Raju.

Description of the work:

The proposed technique for image encryption described in the paper combines SCAN patterns, a carrier image, and the Hill-cipher technique to enhance the security of image transmission over wireless networks. Scan pattern as a crucial component in image encryption algorithm, play a pivotal role in scrambling the pixel arrangement of an image, effectively disorganizing the spatial information, and rendering the encrypted image resistant to statistical attacks.

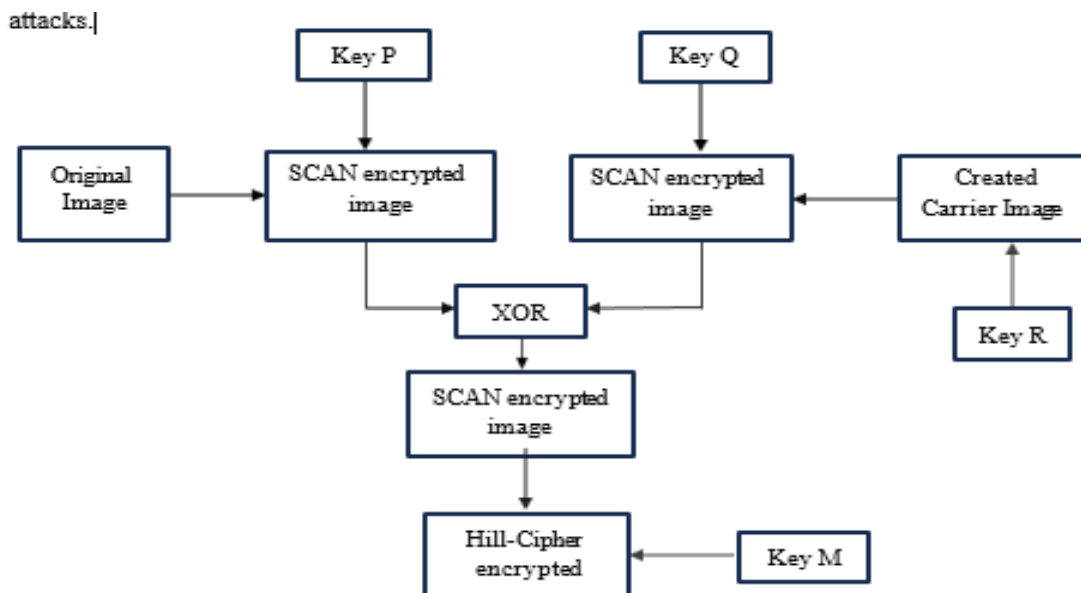


Fig 2.1 : Block diagram for the proposed scheme for Image Encryption using Carrier image, SCAN patterns and Hill-Cipher

The Hill-cipher technique is a cryptographic algorithm that operates on matrices. It involves matrix multiplication with a key matrix to transform the data. In this context, the resultant image from the XOR operation undergoes the Hill-cipher technique, which adds another layer of encryption. The proposed image encryption technique employs a combination of SCAN patterns, a carrier image generated from an ASCII string, and the Hill-cipher technique. This multi-layered approach aims to enhance the security of image transmission over wireless networks, making it more resistant to unauthorized access and decryption [1].

Advantages:

1. Provides multi-layer approach for encryption and enhance the security of the image.
2. Makes the encryption process more resistant to known cryptographic attacks.
3. Adding complexity and making it more challenging for unauthorized users to predict or reverse-engineer the encryption.

Disadvantages:

1. Key management is crucial for the security of any cryptographic system, and the complexity increases with multiple layers of encryption.
2. They may introduce computational overhead, potentially impacting the speed of encryption and decryption.

2.2 “Triple-Layer Image Security Using a Zigzag Embedding pattern”

Authors: Sara Farrag , Wassim Alexan and Hisham

Description of work:

The described work presents a triple-layer message security scheme that combines cryptographic and steganographic techniques to enhance the robustness of hiding secret information within an image and high-security message hiding technique that combines cryptography and steganography. It uses AES-128 encryption for initial protection, followed by chaotic encryption for further enhancement. Finally, it conceals the message within an image using a unique zigzag pattern within the least significant bits of each colour plane. This unpredictable data hiding strategy offers robust security and high capacity, as evidenced by superior image quality metrics and favorable histogram analysis. However, the triple-layered

approach can be computationally expensive and susceptible to advanced steganalysis techniques.

The first two layers employ cryptographic techniques for encryption, beginning with AES-128 encryption on the secret message in the initial layer. Following this, a chaotic logistic map encryption is applied to the output of the first layer to further bolster security. The third layer introduces steganography, utilizing a 2D image steganography technique. Here, the least significant bit (LSB) embedding is executed in a zigzag pattern across the RGB color planes of the cover image. This unique zigzag pattern adds an unpredictable element to the hiding process, enhancing resistance against third-party detection. Notably, the scheme achieves superior performance metrics such as Peak Signal to Noise Ratio (PPSNR), Mean Square Error (MSE), Structural Similarity Index Metric (SSIM), Normal Cross Correlation (NCC), and Image Fidelity (IF) compared to existing literature counterparts. Furthermore, histogram analysis and the attained high capacity underscore the reliability and effectiveness of this steganographic scheme. Overall, this scheme presents a comprehensive and innovative approach to message security, integrating cryptographic and steganographic layers to ensure robust protection with heightened performance and capacity.

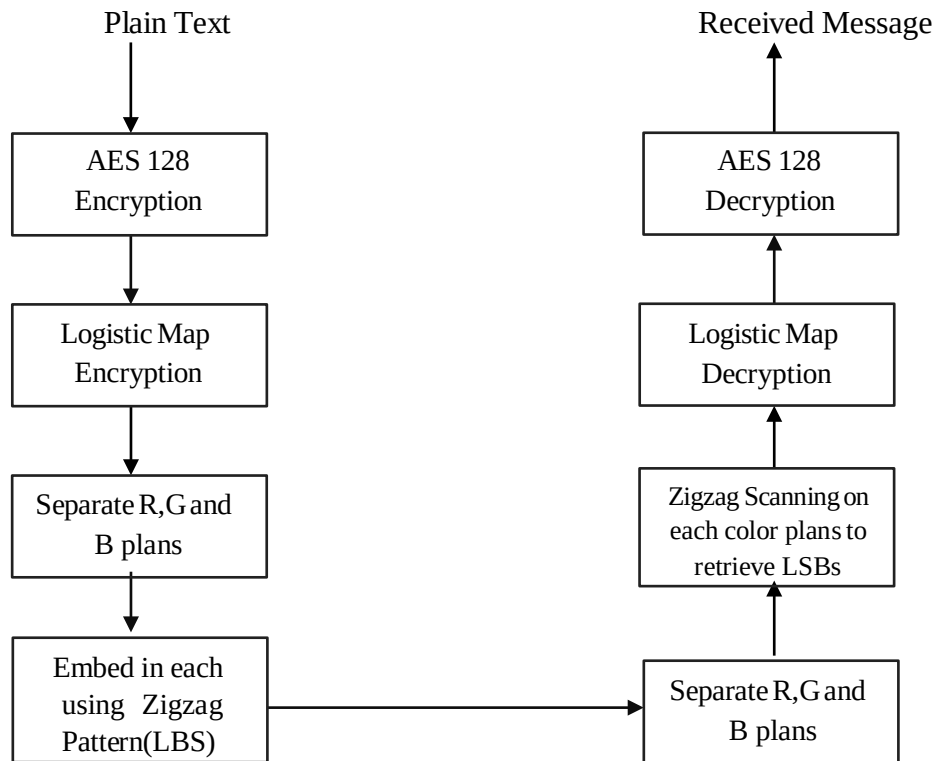


Fig 2.2 : Flow chart illustrating the proposed triple – layer scheme

Image steganography schemes include several ways for embedding secret data into a cover image such as: spread spectrum-based steganography, transform domain-based steganography, model –based steganography and spatial domain-based steganography which in turn divided to pixel value differencing and least significant bit (LSB) substitution steganography approaches [2]. The proposed scheme is composed of three security layers as follows: AES–128 encryption, chaotic logistic map encryption and LSB steganography [2].

Advantages:

1. The triple layer approach offers robust security by combining cryptographic and steganographic techniques.
2. The zigzag pattern makes it difficult for attackers to detect and extract hidden data.
3. The steganographic process maintains high image quality, making it difficult to detect any changes

Disadvantages:

1. The triple-layer approach can be computationally expensive, potentially impacting performance.
2. The scheme might not be suitable for real-time applications due to its computational cost.

2.3 “Image encryption and Decryption Using Scan Pattern and XTEA Encryption algorithm”

Authors: Krishna Raj A, Sharath Chandra N R.

Description of Work:

This project leverages the combined power of SCAN patterns and the Extended Tiny Encryption Algorithm (XTEA) to achieve secure image transmission. SCAN patterns, which rearrange image pixels according to a specific scheme, contribute to enhanced security. Subsequently, the rearranged data undergoes encryption using the XTEA algorithm, a robust and efficient method with a larger key size compared to the standard Tiny Encryption Algorithm (TEA). The use of SCAN patterns has the potential to accelerate both encryption and decryption processes, potentially offering improved performance compared to alternative

techniques. While the secrecy of the XTEA key remains crucial for the overall security of the system, the project presents a promising solution for secure image transmission. By combining SCAN patterns with the XTEA algorithm, this method offers enhanced security.

Various encryption algorithms exist, each tailored to factors such as data size, security requirements, and processing speed. The process involves first rearranging the pixels of an image using SCAN patterns, which serves as a form of preprocessing to obscure the image's content. Then, the rearranged image is encrypted using the Extended TEA algorithm. This project specifically focuses on enhancing network security through image encryption and decryption techniques. By combining SCAN patterns for pixel rearrangement and the Extended TEA algorithm for encryption and decryption, this project aims to provide a robust method for securing images, which can be crucial for protecting sensitive information in various applications, from digital communication to data storage. It employs SCAN patterns to rearrange image pixels, followed by encryption and decryption using the Extended Tiny Encryption Algorithm. By combining these methods, the project aims to provide a robust solution for securing images, ensuring confidentiality and integrity in the face of evolving cyber threats.

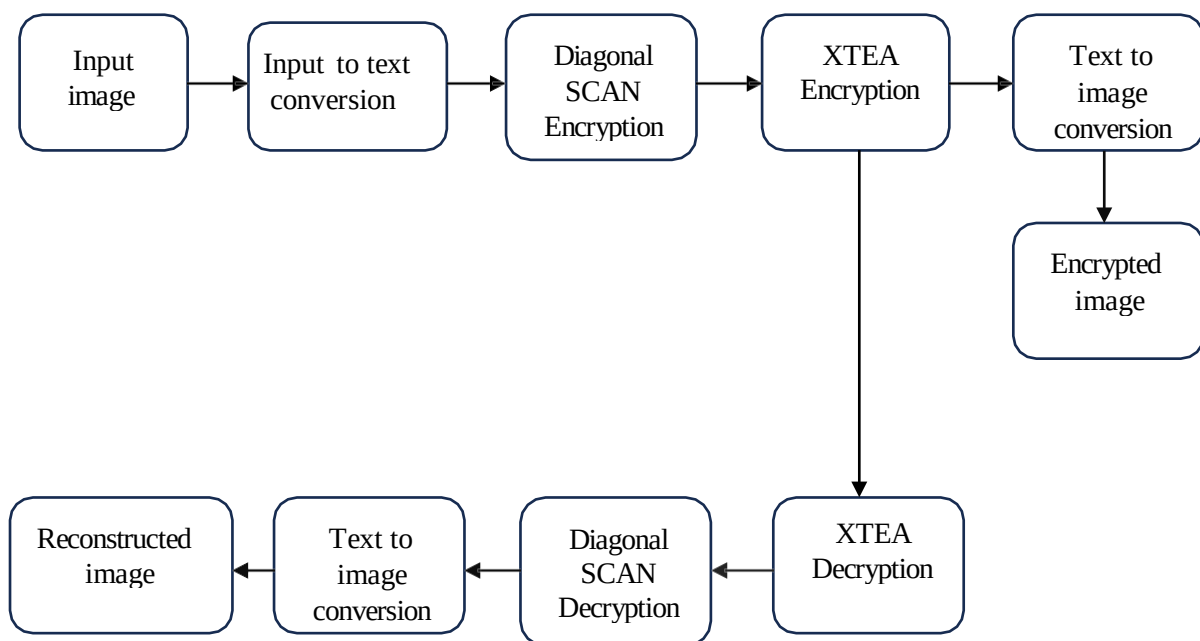


Fig 2.3 : Image encryption and Decryption Using Scan Pattern and XTEA Encryption algorithm Proposed system

Advantages:

1. This encryption system doubles down on security by scrambling pixels twice, making it super tough to crack without the key and pattern.
2. It is also efficient, easy to implement, and scales well for all your image types and sizes.

Disadvantages:

1. The diagonal scan pattern can introduce distortion into the image, making it difficult to discern the original image.
2. The XTEA algorithm has been broken in the past, and new attacks may be developed in the future that could compromise the security of this system.

2.4 “An Image encryption Technique using Logistic map and Z-order curve”

Authors: Shana K U and Anuj Mohamed.

Description of work:

The described work proposes a novel image encryption technique specifically designed for grayscale images, utilizing the Logistic map and Z-order curve and focuses on enhancing the security of grey scale images through a novel encryption technique that combines the chaotic properties of the Logistic map and the spatial scrambling capability of the Z-order curve. The key feature of this method is the generation of a key stream that depends not only on a chaotic map (Logistic map) but also on the characteristics of the original image. The process involves scrambling the original image using a Z-order curve, followed by encryption using a random matrix generated through the Logistic map. [4] suggests that the experimental results and security analysis demonstrate the effectiveness of the proposed method, asserting its high security and resilience against various types of attacks. The key innovation lies in the generation of a key stream that not only relies on the chaotic map but also incorporates information from the original image, thereby adding an extra layer of complexity and security. The proposed technique is a symmetric one, in such a manner that the source and the destination use the same secret key [4].

Algorithm for Encryption process:

Step1: Read an image I of size $M1 \times N1$ where $M1$ is the width and $N1$ is the height of the image. Let $I(m,n)$ denote each pixel with $m=1,2,3,\dots,M1$ and $n=1,2,3,\dots,N1$.

$$y_{k+1} = \mu y_k(1 - y_k) \quad (1)$$

Step 2: Perform pixel scrambling to ensure confusion technique using SCAN method. For pixel scrambling Z-order scan curve is used. Fig.3 shows the scan path of a 4×4 Z-order curve starting at bottom left corner of the matrix. Let the scrambled image can be denoted as SI . Fig.4 shows original image (I) and scrambled image (SI) of Lena using Z-order curve.

Step 3: Generate the secret key which is used by both source and destination. It consists of the following. a. The control parameter of logistic map μ , whose value can be assigned as 3.64. b. The parameter λ with a value as 1021. It can be calculated by trial-and-error process. c. The seed value of logistic map Y_k is calculated using the given equations.

$$T1 = (Tot_S/M1) \bmod 256 \quad (2)$$

$$T2 = (Tot_S/T1) \bmod 256 \quad (3)$$

$$Y_k = T1/T2 \quad (4)$$

Step 4: Using the chaotic Logistic map in (1) generate pseudo-random numbers. Generate random values using the above chaotic map ($M1 \times N1$) +500 times and take the final $M1 \times N1$ random numbers. Then perform the following calculations on the y_i random values.

$$x(i) = \text{abs}(\lfloor y_i * y_i * y_i * \beta \rfloor) \bmod M1 \quad (5)$$

Here i may range from 1, 2,..... $M1 \times N1$. Where μ is taken as 3.64 for finding the y_i values and

β is a parameter of value 123456789. Construct a matrix (SRM) of size $M1 \times N1$ with these $x(i)$ values.

Step5: Perform the following operations for completing the encryption.

$$Tot_S = \sum_{j=1}^N \sum_{i=1}^M SI(i, j) \quad (6)$$

For each pixel (i,j) calculate the following,

$$SP(i, j) = (TotS + SI(i, j) + \sum_{i=1}^M rsum(i, j) + \sum_{j=1}^N csum(i, j) + \lambda) \bmod 256 \quad (7)$$

Step 6: Encrypt the image using (8) to obtain the cipher image E_img .

$$E_img(i, j) = Sp(i, j) \oplus SRM(i, j) \quad (8)$$

Algorithm for Decryption process:

In the proposed technique the secret key used for decryption process is also the same as encryption. In decryption process the different steps performed in encryption are applied in reverse manner. Use (9) to obtain the intermediate image Sp and on this image perform inverse Z-order scan strategy.

$$Sp(i, j) = E_img(i, j) \oplus SRM(i, j) \quad (9)$$

Advantages:

1. By incorporating both the chaotic nature of the Logistic map and the spatial diffusion provided by the Z-order curve, the encryption scheme enhances the overall security of the system.
2. The dependence of the key stream on both the chaotic map and the original image contributes to a more robust encryption process.

Disadvantages:

1. The proposed method may not be immune to sophisticated adversarial attacks, especially if the underlying cryptographic primitives are not carefully chosen or if there are vulnerabilities in the implementation.
2. Adversarial machine learning techniques could potentially exploit weaknesses in the encryption scheme.

2.5 “An Image Cryptography using Henon Map and Arnold Cat Map.”

Authors : Pranjali Sankhe, Anita Lahan.

Description of the work:

The work introduces a new symmetric image encryption algorithm aimed at securing digital data, especially multimedia content such as text, images, and videos. Cryptography is the process of encryption and decryption of the data and used for making data, images secure

and confidential. There is also another side where the attackers, unauthorized user gets an opportunity to reuse, retrieve, disturb the images. In this paper we will apply the chaotic encryption and chaotic decryption on an image. Chaotic Algorithm is a well-used method in real time secure image transmission system. The proposed method leverages Henon's chaotic system and a novel approach involving pixel shuffling to achieve effective and efficient image encryption. Additionally, the Arnold Cat Map, a discrete system with specific phase space characteristics, is employed to further enhance the encryption process. Utilizes Henon's chaotic system as a foundational element for generating chaotic behaviour. Chaotic systems are known for their sensitivity to initial conditions, contributing to randomness in the encryption process. Incorporates the Arnold Cat Map, a discrete dynamical system, which transforms pixel coordinates and alters the distribution of pixel values. This process aims to provide additional security by stretching and folding trajectories in phase space.

The proposed method enhances the security and efficiency of image encryption by utilizing the Arnold Cat Map for trajectory stretching and folding in phase space, providing a robust defence against unauthorized access and potential attacks in the digital transmission of multimedia data [5].

Information security has become an important issue with the rapid development of communication networks. Since large number of digital images are transmitted over the networks, it is necessary to ensure the confidentiality of images to protect them against unauthorized access. Image security is very important in almost all areas like military affairs, medical field, and confidential business.

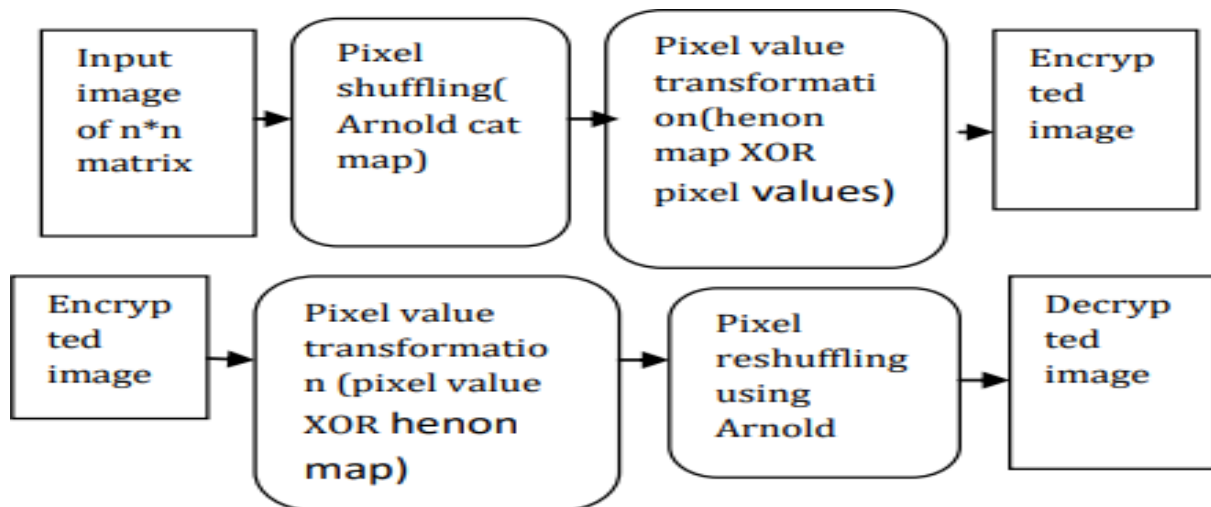


Fig 2.4 : An Image Cryptography using Henon Map and Arnold Cat Map Proposed system

Multimedia is defined as the field that deals with different forms of information such as text, images, audio, and videos in an integrated fashion. Now-a-days digital images are used frequently for communication [5].

Advantages:

1. This innovative approach enhances data confidentiality.
2. Provides a robust defense against both active and passive attacks in digital image transmission.

Disadvantages:

1. The complexity introduced by chaotic algorithms may present a challenge in terms of a computational overhead, potentially impacting the efficiency the encryption process.

2.6 “Chaos-Based Confusion and Diffusion of Image Pixels Using Dynamic Substitution”

Authors: Abdhulla Qayyum

Description of the work:

This work discusses the growing importance of encryption in wireless and mobile communication, particularly in the face of increasing cyber threats. It introduces a novel image encryption algorithm leveraging chaos theory and dynamic substitution, based on two-dimensional Henon and Ikeda chaotic maps, along with a substitution box (S-box) transformation. The proposed scheme ensures enhanced security and robustness, as validated through various security tests such as information entropy, histogram investigation, correlation analysis, energy, homogeneity, mean square error, and key space analysis. Simulation results and comparisons with state-of-the-art algorithms demonstrate the efficiency and security of the proposed encryption scheme, as evidenced by superior entropy values, a large key space is 10^{244} [6].

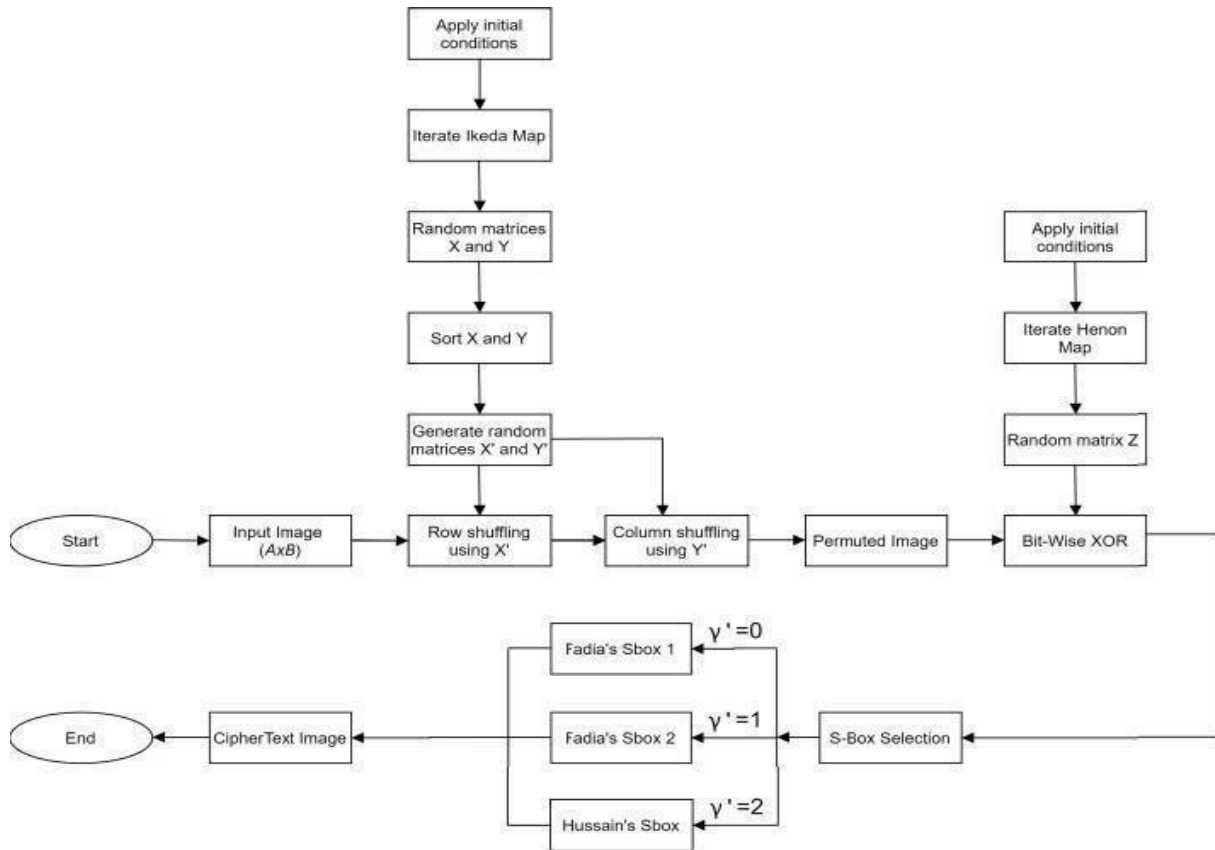


Fig 2.5 : Chaos-Based Confusion and Diffusion of Image Pixels Using Dynamic Substitution proposed system

2.7 “Image Encryption Using Chaotic Map And Related Analysis”

Authors: Diya Achu Pradeep, Harsha.

Description of the work:

This work discusses the growing significance of data transmission over the internet, particularly in critical fields such as Banking, Military, and Medicine, where reliance on telemedicine has increased due to travel limitations. Recognizing the vulnerability of these fields to intruders, the paper proposes the use of chaos encryption, a method known for its enhanced and conducting a security analysis to demonstrate the algorithm's robustness. The research also explores various chaotic map combinations, revealing a novel map combination that enhances the algorithm's effectiveness, as supported by simulation results, affirming the proposed method's simplicity and robustness for practical applications in secure image

transmission. In this work, first the encryption and decryption using the original version of henon chaotic map is done. Followed by which an analysis of the same under different noise conditions are analysed. After this, different versions of henon map are reviewed. But it is always desired to find a better chaotic combination for this application. So, for that a combination of these three single chaotic maps are defined and simulated [7].

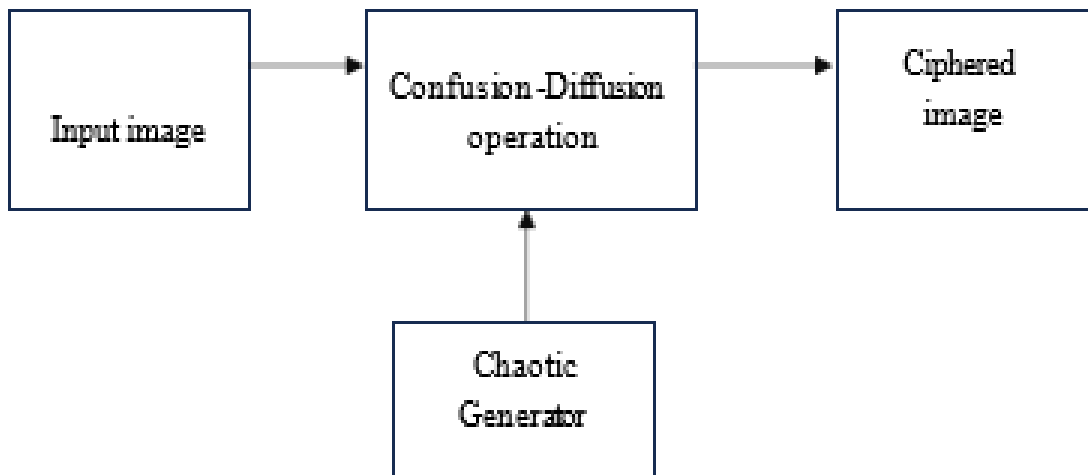


Fig 2.6 : Encryption phase of Image Encryption Using Chaotic Map And Related Analysis

Advantages:

1. This method provides robust protection against intruders, addressing the growing need for secure data transmission, especially in telemedicine where the confidentiality of medical information is paramount.
2. Ensures the protection of graphical image content exchanged over the internet.

Disadvantages:

1. They is a complex implementation procedure and the need for specialized expertise.
2. Computational complexity, which could hinder real-time processing, and their vulnerable to compromise if underlying mathematical structures.

2.8 “Exploring the Efficacy of Image Encryption using Independent, Combined and Improved Chaotic Maps”

Authors: C Manikandan, S Shreela, S Karunyaa , Priya Selvam , A Rajesh.

Description of the work:

The rapid advancement of digital communication and data storage technologies has heightened the need for secure preservation of sensitive information in today's rapidly evolving digital communication and data storage environment, this study investigates the effectiveness of image encryption methods utilizing chaotic maps. The research explores one-dimensional (1D), combined, and enhanced chaotic maps, revealing that the Logistic, Tent, and Sine maps exhibit superior performance in correlation, differential, histogram, and entropy analyses across all three domains, establishing their efficacy in ensuring the security and integrity of visual data through image encryption. Image encryption plays a vital role in safeguarding the security and integrity of visual data, with chaotic maps, specifically the Logistic, Tent, and Sine maps, demonstrating superior effectiveness in correlation, differential, histogram, and entropy analysis across one dimensional(1D), combined, and enhanced domains. The quality and strength of encryption are compared for all 1D chaotic, hybrid, and improved maps. 1D chaotic maps are a helpful tool. Their simplicity and complex behaviour make them adaptable and widely valuable for various scientific, engineering, and computing domains [8].

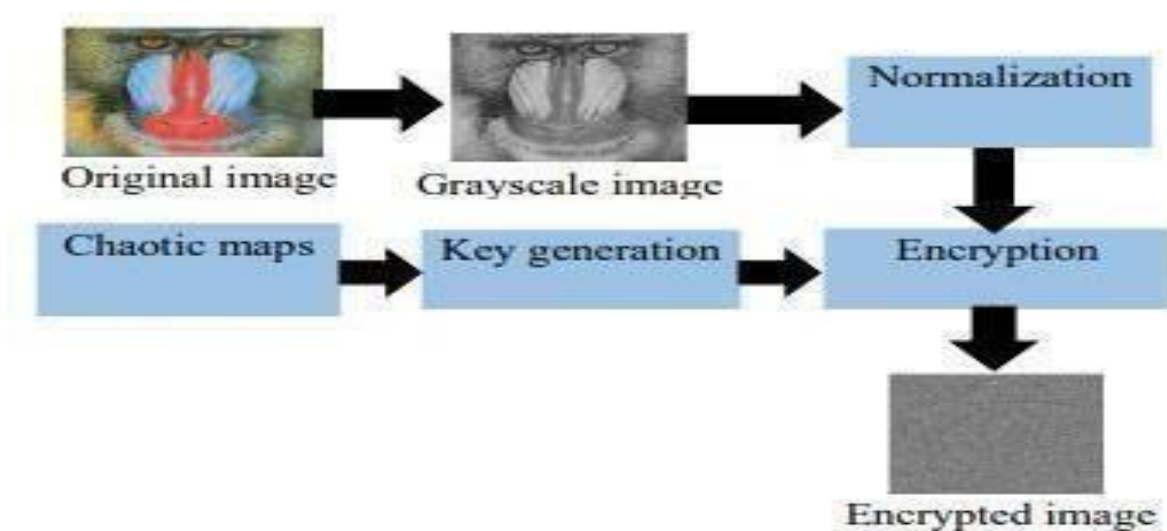


Fig 2.7 : Exploring the Efficacy of Image Encryption using Independent, Combined and Improved Chaotic Maps

Advantages:

1. It provides a robust and highly effective means of ensuring the security and integrity of visual data in the rapidly evolving landscape of digital communication and data storage technologies.
2. Maintains favourable statistical characteristics in the encrypted data.

Disadvantages:

1. This complexity could impact the efficiency of the encryption process, making it less suitable for applications with limited computational resources.
2. It reduces the efficiency of the encryption process.

2.9 “Chaos Based Image Encryption using Arnold’s cat map confusion and Henon Map diffusion”

Authors: Anak Agung Putri Ratna, Frenzel Timothy Surya, Diyanatul Husna, IKetut Eddy Purnama, Ingrid Nurtanio, Afif Nurul Hidayati, Mauridhi Hery Purnomo, Supeno Mardi Susiki Nugroho, Reza Fuad Rachmadi.

Description of the work:

This work introduces a specialized image encryption system designed for securing tele-dermatology data, specifically focusing on skin disease images, with the encryption and decryption operations taking place on the client side. The system integrates chaos- based encryption techniques, employing both confusion and diffusion methods, utilizing Arnold's cat map for confusion and the Henon map for diffusion. The chaotic maps are initialized with a 30- digit secret key generated through Diffie–Hellman key exchange. Precision parameters are set, with the Henon map values at 10^{-14} , and varying p and q values for Arnold's cat map during generation. Experimental results showcase that encrypted images exhibit uniformly distributed histograms across gray values, and key sensitivity tests emphasize a significant impact on similarity indices with slight variations in the secret key. The decryption process generates another chaotic image, demonstrating average entropies and highlighting the system's effectiveness in maintaining data security for tele-dermatology applications [9].

This section will discuss the steps involved in the proposed encryption scheme. To achieve the properties of confusion and diffusion, two different two-dimensional iterative chaotic maps are employed [9].

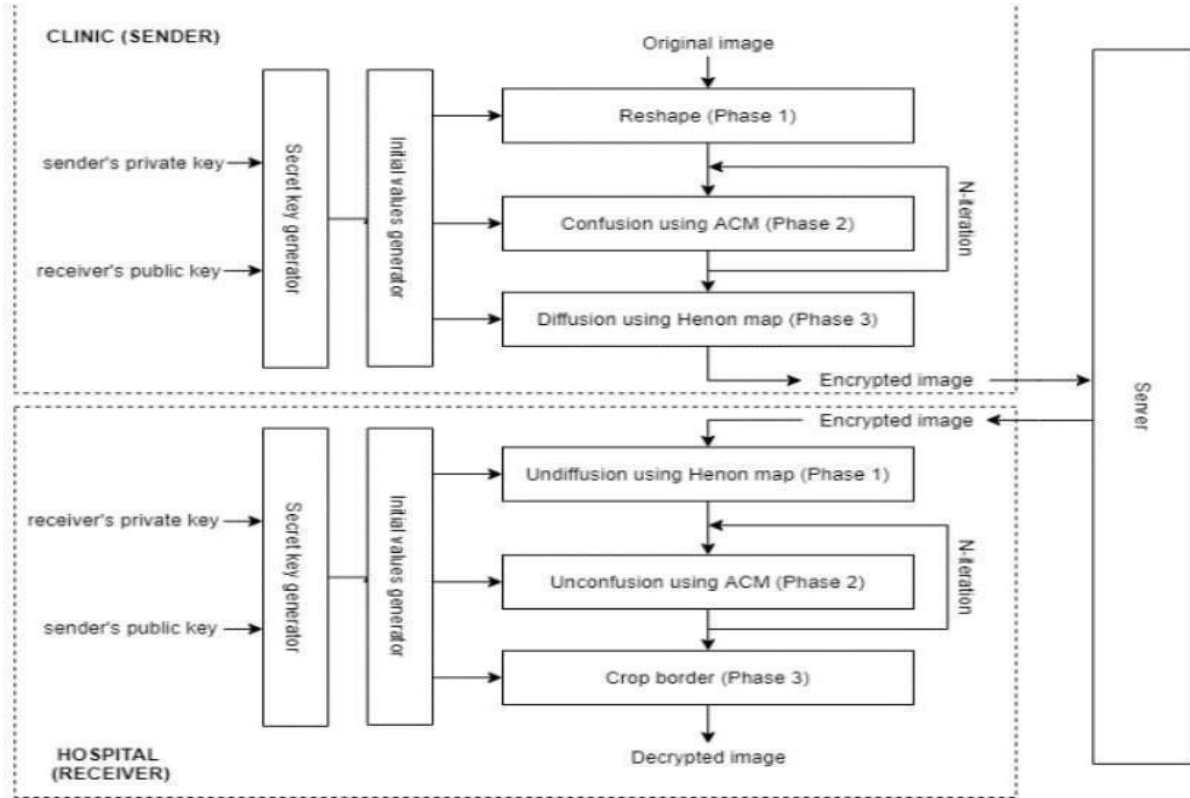


Fig 2.8 : Proposed scheme of Chaos Based Image Encryption using Arnold's cat map confusion and Henon Map diffusion

Advantages:

1. Provides strong security for tele-dermatology data due to the utilization of chaos-based encryption with high sensitivity to key differences and resistance to decryption without the correct secret key.
2. Ensures robust protection for sensitive skin disease images with the encryption and decryption processes executed on the client side.

Disadvantages:

1. The computational complexity of the encryption and decryption process, which may impact real-time application performance.

2.10 “Self Adaptive Image Encryption Based on Orthogonal Scan Patterns Using Arnold Cat Map and Tent Map”

Authors: Rashmi P and Supriya M C.

Description of the work:

In response to the increasing reliance on social media and the consequential need for secure image processing and network communication, this article introduces a novel hybrid image encryption technique. The method incorporates Continuous Orthogonal Scan, Arnold Cat Map, and Tent Map for image block partitioning to address limitations observed in existing encryption methods, such as susceptibility to plaintext attacks, simplistic algorithm structures, and small key spaces. The proposed technique utilizes continuous orthogonal scan to rearrange pixels within each block, employs Arnold cat map to introduce randomness in start-points, and utilizes Tent map to generate secret key based on image content, forming a random matrix [10]. The three main steps of the proposed algorithm consist of XOR operation, Continuous orthogonal scan and non-overlapping block partition as shown in Figure 2.9. Under the secret key control, non-overlapping blocks are obtained by dividing the input image. The order of encrypted image blocks is identified by using secret key k1. The random start-points are generated for continuous orthogonal scans by using Arnold cat map [19,20] for improving the security [10].

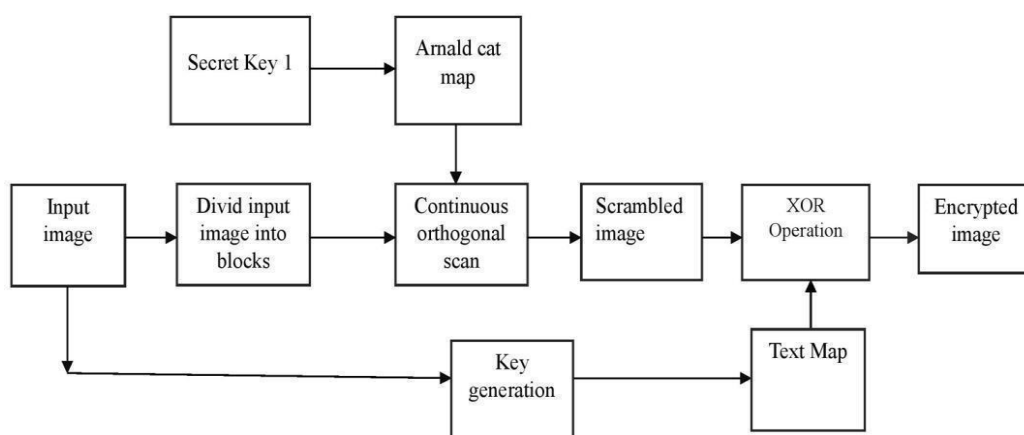


Fig 2.9 : Proposed scheme of Self Adaptive Image Encryption Based on Orthogonal Scan Patterns Using Arnold Cat Map and Tent Map

Advantages:

1. Provides heightened security through a complex algorithm structure and increased key space, effectively overcoming limitations in existing methods.
2. Its adaptability to diverse image content and demonstrated effectiveness in a simulation experiment underscore its potential as a robust solution for safeguarding user image data in the context of escalating social media usage.

Disadvantages:

1. Encryption technique may increase computational complexity, which could potentially impact processing speed and resource efficiency.

2.11 “A Survey on Image Encryption using Chaos-based Techniques”

Authors: Veena G, Dr. Ramakrishna M.

Description of work:

The work is on image encryption using chaos-based techniques. It discusses the limitations of traditional encryption methods for images and highlights the suitability of chaos-based techniques due to their properties such as sensitivity to initial conditions, pseudorandom number generation, ergodicity, and density of periodic orbits. The survey focuses on various chaotic maps like the logistic map, piecewise linear chaotic map (PWLCM), tent map, 3D cat map, Bülban chaotic map, and hybrid chaotic map for image encryption. To secure information present in the image from unauthorized user access during transmission and storage, image encryption is used. Image encryption can be achieved by changing pixel positions as well as changing contents of pixels as a result the true content of the image is completely changed [11]. Sensitivity to initial condition is one property of chaos-based technique in which a minute variation in initial conditions will result in a large variation in behaviour of the system [11].

Following steps are followed to encrypt the image:

1. Key generation: 128-bit sequence number is split into kpx, kpy, kpz, kqx, kqy, kqz that are used in 3D cat map and kl, ks are used in logistic map.

2. Conversion of the 2D image into 3D: The image is split to form several three-dimensional cubes.
3. Apply 3D cat map on the result of step b.

Advantages:

1. The work aims to identify the best chaotic map for image encryption by comparing various chaotic maps such as the logistic map, piecewise linear chaotic map (PWLCM), tent map, and others. This helps in understanding the strengths and weaknesses of each method.
2. It discusses the resistance of different encryption techniques to various attacks, such as statistical attacks, differential attacks, and brute-force attacks. This provides insights into the robustness of each method.

Disadvantages:

1. The speed performance of some of the proposed algorithms is not ideal compared to other algorithms. For example, in the case of DNA encoding combined with chaos-based logistic map, the speed performance is not optimal compared to other algorithms.

CHAPTER 3

SYSTEM DESIGN AND IMPLEMENTATION

To ensure the safe transmission of the image information in communication and improve the security performance of image encryption algorithm, a hybrid chaotic encryption algorithm is proposed with higher security based on chaotic system. The traditional encryption algorithms such as DES, AES and RSA cannot achieve ideal results, chaotic encryption schemes have attracted the researchers' attentions. Chaotic systems have obvious advantages in secure communication. In one hand, chaotic signals have the advantages of pseudo-randomness, ergodicity, topological transitivity and extreme sensitivity to initial values and parameters, which can be used in the process of cryptographic permutation and diffusion. On the other hand, the chaotic sequence can be generated by a simple iterative method, and the encryption is good in real time. Therefore, applying chaos to image encryption can improve the security of the encryption system.

3.1 SYSTEM DESIGN

Here, a hybrid chaotic encryption algorithm is proposed with higher security based on chaotic system. A chaos based crypto system improves the security of the encryption system and ensure the safe transmission of the image information in communication. The key generation part is included as a separate process to enhance the privacy.

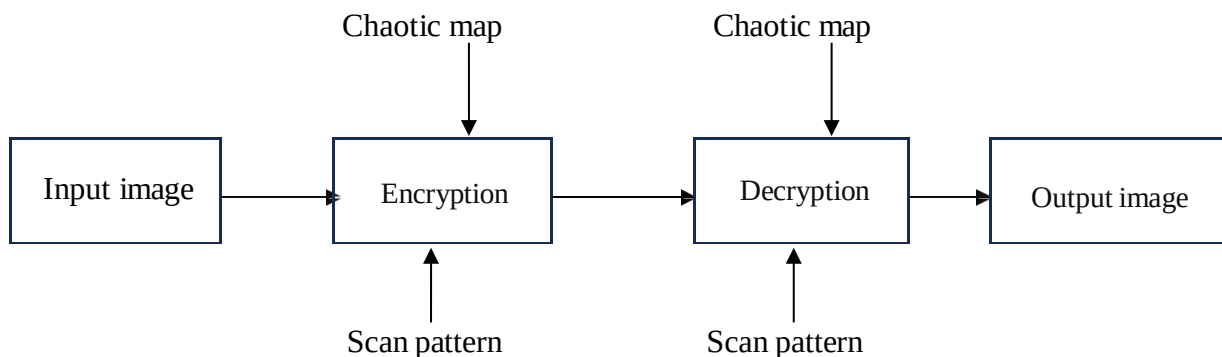


Fig 3.1: Chaos based encryption

As shown in Fig 3.1, the input image is given as input for the encryption process. The image is encrypted with the sequence generated by the chaotic map. Once the message is received, to decrypt the encrypted message, we use the key which is the sequence generated by the Arnold's Cat map and Logistic map which retrieve the original or output image.

The major advantage of chaotic maps are confusion and diffusion where confusion means that each binary digit (bit) of the ciphertext should depend on several parts of the key, obscuring the connections between the two. The property of confusion hides the relationship between the ciphertext and the key. This property makes it difficult to find the key from the ciphertext and if a single bit in a key is changed, the calculation of most or all the bits in the ciphertext will be affected. Confusion increases the ambiguity of ciphertext and it is used by both block and stream ciphers [12]. Diffusion means that if we change a single bit of the plaintext, then about half of the bits in the ciphertext should change, and similarly, if we change one bit of the ciphertext, then about half of the plaintext bits should change. This is equivalent to the expectation that encryption schemes exhibit an avalanche effect. The purpose of diffusion is to hide the statistical relationship between the ciphertext and the plain text.

3.1.1 Encryption

An image encryption scheme is designed in this work by using scan pattern and chaotic maps such as Logistic map and Arnold's cat map. The proposed encryption algorithm consists of three stages such as scan operation, confusion, diffusion, generation of key image and diffusion [1].

The Fig 3.2 shows the encryption algorithm. Here, the Lena image of size 256x256 is taken as input image. The encryption process begins with the selection of the original image in .png format. A scan operation is then performed on the input image utilizing the Zeta Z scan pattern, which utilizes the image's height and width dimensions. Following this, the Confusion operation is applied to the scanned image using Arnold's cat map, renowned for its chaotic nature, to introduce complexity and disorder. Random key sequences are generated using the Logistics map, contributing further to the unpredictability of the encryption.

Subsequently, the Diffusion operation is executed by shuffling the pixels of the image using the Spiral-In scan pattern, which also exhibits chaotic behaviour, effectively dispersing

the image's information. The encryption process culminates with the XOR operation being performed between the pixel values generated from the Spiral-In scan pattern and the random key sequence generated by the Logistics map. This procedure results in the creation of the Cipher image or Encrypted image, thereby completing the encryption process and ensuring the confidentiality and security of the original image.

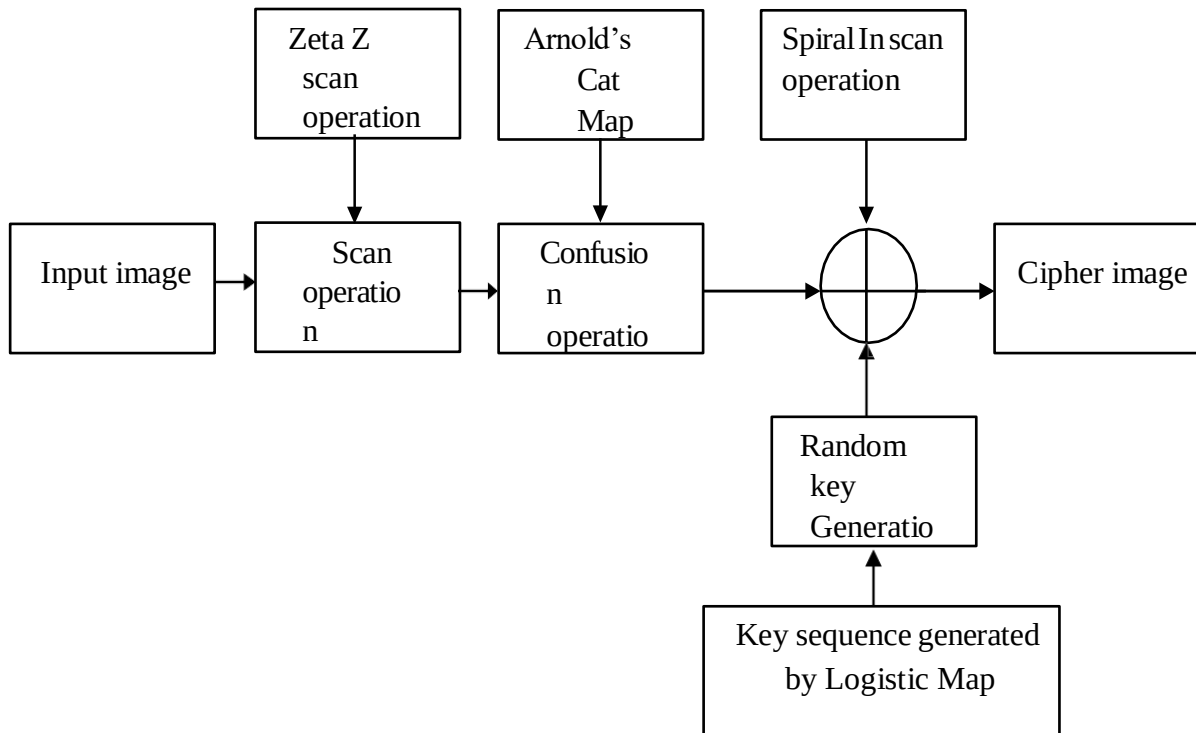


Fig 3.2: Block diagram for the proposed encryption algorithm

3.1.2 Decryption

The obtained encrypted image is used for decryption algorithm. In decryption the exact same steps involved in encryption is reversed to get the original image. The below Fig 3.3 shows the decryption algorithm. Here the cipher image is considered as the input image, the Diffusion operation is initiated, involving the rearrangement of pixels within the cipher image using the Spiral In scan pattern, which exhibits chaotic behaviour. Subsequently, a Random key sequence is generated utilizing the Logistics map, renowned for its chaotic properties.

This key sequence is then used in conjunction with the XOR operation, applied between the pixel values derived from the Spiral-In scan pattern and the generated key sequence, effectively reconstructing the original image. Following this step, the Confusion operation is executed on the image utilizing Arnold's cat map, known for its chaotic nature, further enhancing the decryption process. To finalize the retrieval of the original image, the Zeta Z scan pattern operation is employed, meticulously scanning, and reconstructing the image's original form. Ultimately, the decryption concludes with the successful recovery of the original image from the cipher image, marking the culmination of the decryption process.

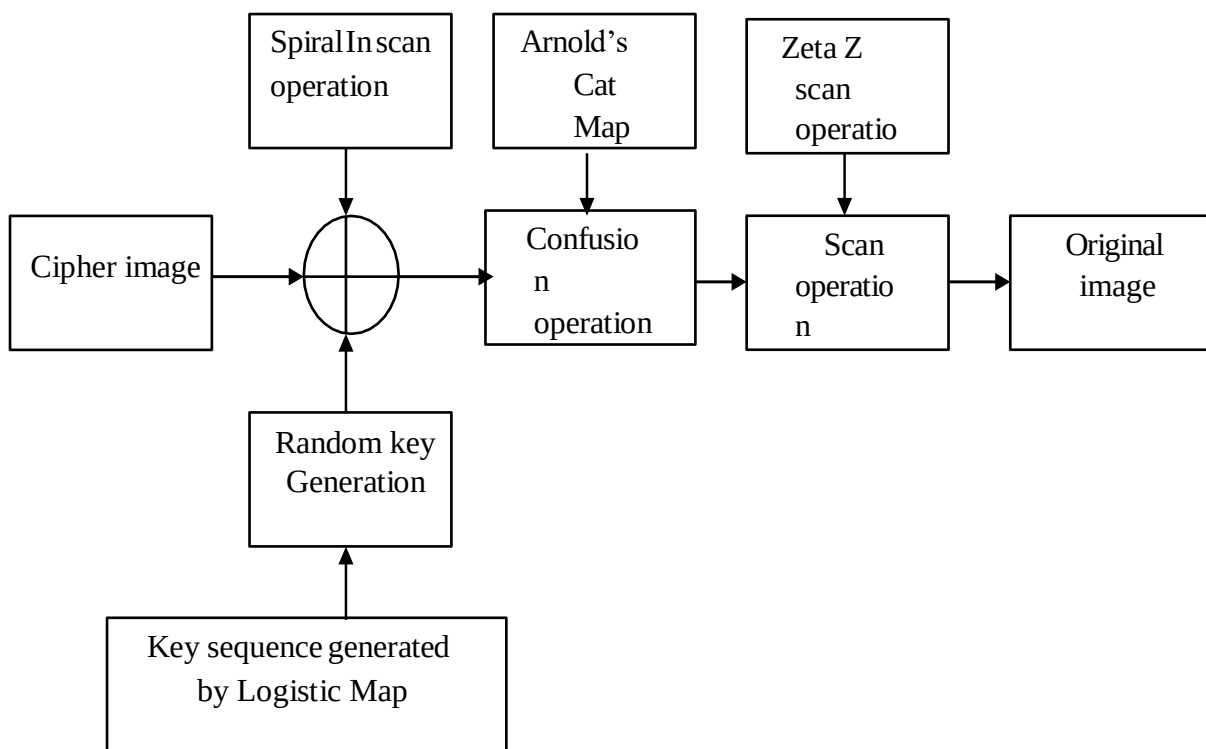


Fig 3.3 : Block diagram for the proposed decryption algorithm

3.2 IMPLEMENTATION

Creating a secured, flexible, and robust communication system the chaos based crypto system is built. The major advantage of scan pattern and chaotic maps are confusion and diffusion in this work Logistic map and Arnold's cat map are used. Initially, the encryption is done where the original image is converted into the cipher image and followed by the decryption process which is exactly reverse as the encryption to obtain the original image.

3.2.1 ENCRYPTION

An image encryption scheme is designed in this work by using scan pattern and chaotic maps such as Logistic map and Arnold's cat map. The proposed encryption algorithm consists of three stages such as scan operation, confusion, diffusion, generation of key image and diffusion.

ENCRYPTION ALGORITHM:

STEP 1: The original image in .png format is selected for the encryption process.

STEP 2: The scan operation is executed on the input image using the Zeta Z scan pattern, utilizing the image dimensions, namely height and width.

STEP 3: The Confusion operation is applied to the image obtained from the scan operation using Arnold's cat map, known for its chaotic nature.

STEP 4: Random key sequences are generated using the Logistics map.

STEP 5: The Diffusion operation is carried out by shuffling the pixels of the image using the Spiral-In scan pattern, which also exhibits chaotic behaviour.

STEP 6: The XOR operation is performed between the pixel values generated from the SpiralIn scan pattern and the random key sequence generated by the Logistics map.

STEP 7: The Cipher image or Encrypted image is successfully created, marking the completion of the encryption process.

3.2.1.1 Scan operation

When applying SCAN to encrypt an input image, the pixels of the image are rearranged using a set of zeta z scan pattern. For the purposes of this work, we assume that the maximum image size is 256×256 , with a partition size of 8×8 blocks uniformly maintained throughout.

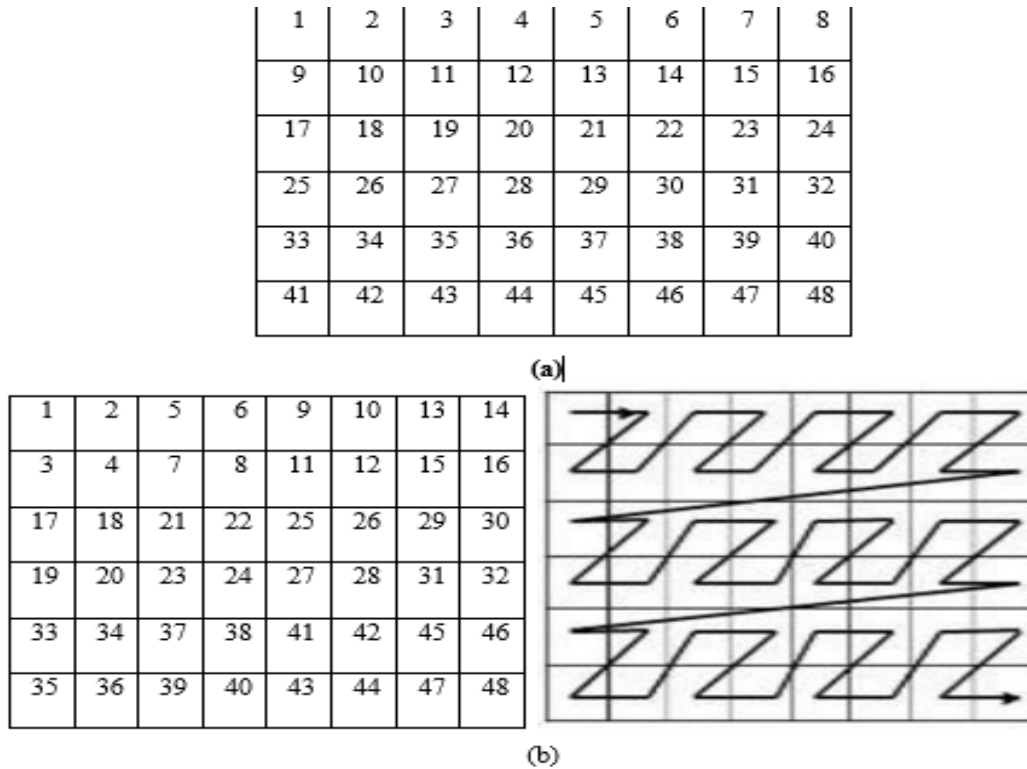


Fig 3.4: (a) Matrix form of a Scan structure, (b) Matrix form for Zeta Z.

In the above process we performed the zeta z scan pattern operation in which pixels are shuffled in zig-zag operation it is shown in the Fig 3.4.

3.2.1.2 Confusion

In this stage, Lena image is considered as input image of size 256×256 which is converted into grayscale, the image pixels are confused by using the x sequence generated by an Arnold's Cat map as shown in equation (2), where the pixels positions are shuffled over the whole image without changing the value of the pixels and the image becomes unidentifiable. Therefore, the initial conditions of Arnold's Cat map serve as the first secret key. The most

important property of Arnold's cat map is that it rearranges the position of image pixel, but after several iterations it returns to the original pixels positions and thereby produces the original image. The Arnold's Cat chaotic map is described as follows

$$\Gamma: (x, y) = (2x + y, x + y) \bmod 1 \quad (1)$$

Where (X_n, Y_n) is the pixel location of the original image, and (X_{n+1}, Y_{n+1}) is the new pixel location in the shuffled image after applying Arnold transformation.

3.2.1.3 Generation of key sequence

To improve the security further, the third stage of the encryption process aims to generate encryption key sequence. Where the encryption key sequence is generated by logistic map.

$$X_{n+1} = \mu * X_n(1 - X_n) \quad (2)$$

Where the parameter μ is taken in the range [3.57 to 4]. Here, the initial value X_0 and the parameter μ are regarded as the secret keys

3.2.1.4 Diffusion

In the diffusion process, where the confused image is shuffled using Spiral In scan operation and XORed with encryption key sequence to produce the cipher image.

3.2.2 DECRYPTION

The obtained encrypted image is used for decryption algorithm. In decryption process, the exact same steps involved in encryption is reversed to get the original image. The decryption algorithm is given below.

DECRYPTION ALGORITHM:

STEP 1: The cipher image obtained from the encryption process is selected for decryption.

STEP 2: The Diffusion operation is executed by rearranging the pixels of the cipher image using the Spiral In scan pattern, which also demonstrates chaotic behaviour.

STEP 3: A Random key sequence is generated using the Logistics map, known for its chaotic behavior.

STEP 4: The XOR operation is then applied between the pixel values derived from the Spiral-In scan pattern and the random key sequence generated by the Logistics map, reconstructing the image from these pixel values.

STEP 5: The Confusion operation is carried out on the image using Arnold's cat map, renowned for its chaotic nature.

STEP 6: The Zeta Z scan pattern operation is employed on the image to retrieve the original image.

STEP 7: The original image is successfully recovered from the cipher image, marking the completion of the decryption process

3.2.2.1 Diffusion

The cipher image is considered as the input image. Then the input image is shuffled using Spiral In scan operation then XORed with the generated encryption key sequence which is generated by logistic map.

3.2.2.2 Generation of key sequence

To retrieve the image further, the second stage of the encryption process aims to generate decryption key sequence. Where the decryption key sequence is generated by logistic map.

3.2.2.3 Confusion

In confusion process, the reverse diffused image is considered as input image. The image pixels are confused by using x sequence generated by Arnold's Cat map as shown in equation (2), where the pixels positions are shuffled over the whole image without changing the value of the pixels and the image becomes unidentifiable. Therefore, the initial conditions of Arnold's Cat map serve as the second secret key. The most important property of Arnold cat map is that it rearranges the position of image pixel, but after several iterations it returns to the original pixels positions and thereby produces the scan image.

3.2.2.4 Scan operation

When applying scan operation to decrypt a scan image, the pixels of the image are rearranged using a zeta z scan operation the original image is retrieved. Once the scan is complete, the collected data is analyzed and processed to reconstruct the original image. This reconstruction is achieved by applying various mathematical techniques, such as interpolation and inverse transformation, to interpolate missing information and correct distortions introduced during scanning. Through the Zeta Z scan pattern operation, the fidelity of the original image can be effectively restored, facilitating accurate image retrieval and analysis in diverse fields.

CHAPTER 4

RESULTS AND ANALYSIS

The Arnold's cat map, Logistic map sequences are generated. By the process of scan operation, confusion, generation of encrypted key sequence and diffusion process, the cipher image is obtained. After successful decryption of the Cipher image the original image is obtained, the GUI is built in Python API and integrated as front end using the Tkinter module. The result analysis to evaluate the chaos based crypto system is done in terms of correlation coefficient analysis, MSE & PSNR, NPCR &UACI, SSIM and finding the information entropy.

4.1 RESULTS

The encryption and decryption algorithm are simulated with software Python API with a python version 3.12 and the results along with analysis are as follows.

4.1.1 Encryption Results

An image encryption algorithm should be able to encrypt different types of plain images into random-like cipher images. Meanwhile, only with the correct key, the original images can be correctly encrypted. Lena image of size 512x512 is considered as input image as shown in Fig 4.1.



Fig 4.1 : Input Image

The input image is considered as input for the confusion process. The image is shuffled Using the zeta z scan pattern. The obtained scan image is as shown in Fig 4.2.

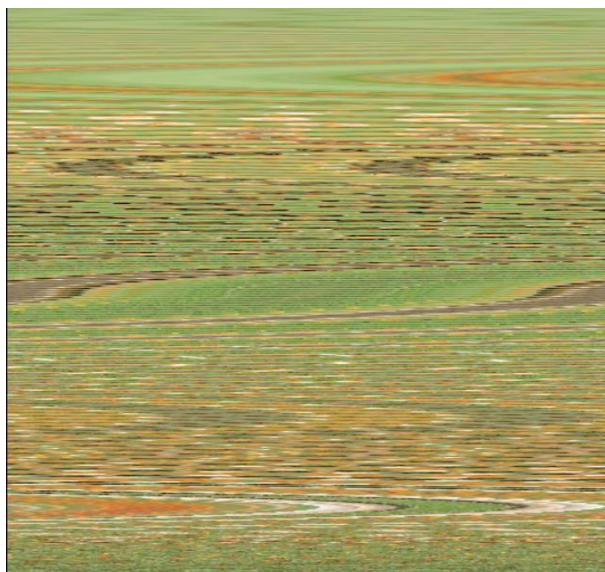


Fig 4.2 : Scan image

The scan image is considered as input for the confusion process. The image is confused Using the x sequence generated by Arnold's Cat map as shown in equation (2) . The obtained confused image is as shown in Fig 4.3.



Fig 4.3 : Confused Image

A series of random key sequences are first generated through the Logistics map. The Diffusion operation is then executed by shuffling the pixels of the image using the Spiral-In scanpattern, which also exhibits chaotic behaviour, enhancing the security of the encryption. Subsequently, the XOR operation is applied between the pixel values derived from the Spiral- In scan pattern and the random key sequence generated by the Logistics map. As a result, the Cipher image, or Encrypted image, is successfully generated as shown in Fig 4.4

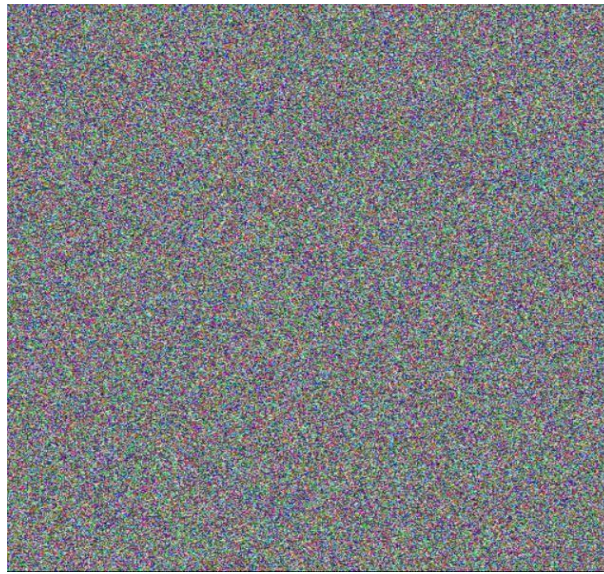


Fig 4.4 : Cipher Image

4.1.1 Decryption Results

An image decryption algorithm should be able to decrypt the cipher image into original image. Only with the correct key, the cipher images can be correctly decrypted. The cipher image is considered as input image as shown in Fig 4.5.



Fig 4.5 : Input cipher Image

The XOR operation is applied between the pixel values obtained from the cipher image by using Spiral-In scan pattern and the random key sequence generated by the Logistics map. As a result, the confused image is successfully generated as shown in Fig 4.6.



Fig 4.6 : Confused Image

The confused image obtained from the diffusion process is confused using the Arnold's cat map operation to obtain the scan pattern image as shown in the Fig 4.2. Then zeta z scan operation is applied to the image to obtain the original image as shown in the Fig 4.7.



Fig 4.7 : Output Image

4.1.2 GUI DESIGN

Tkinter is a popular Python library for creating graphical user interfaces (GUIs). Tkinter's ease of use makes it suitable for both beginners and experienced developers alike. With its robust set of features and cross-platform compatibility, Tkinter remains a go-to choice for developing desktop applications in Python. Layout of the Graphical User Interface as shown in Fig 4.8.

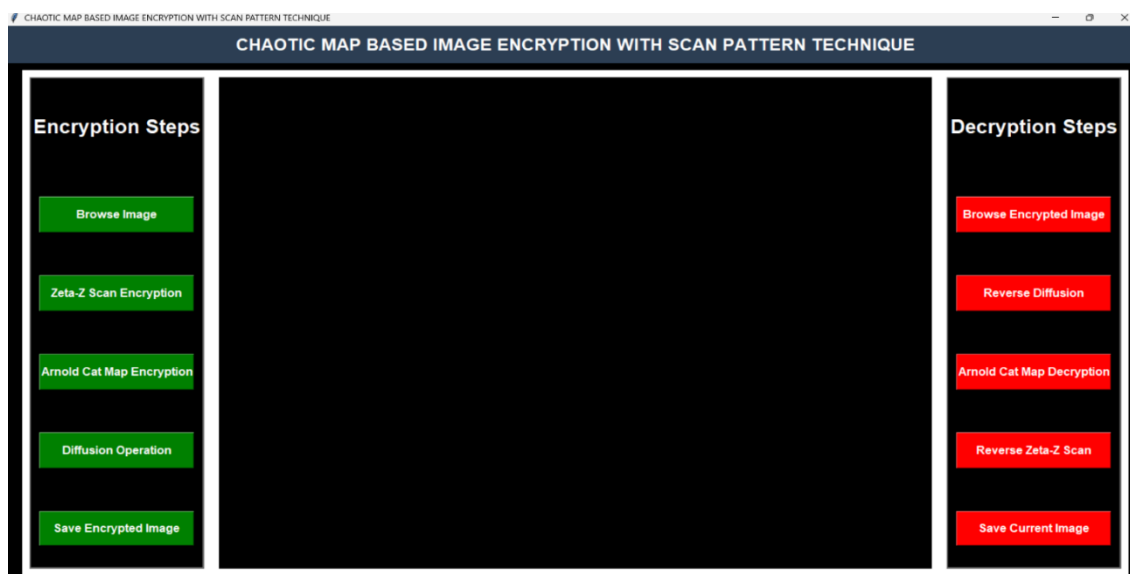


Fig 4.8 : Layout of the GUI

The Browse Button is used to select the input image as shown in Fig 4.9

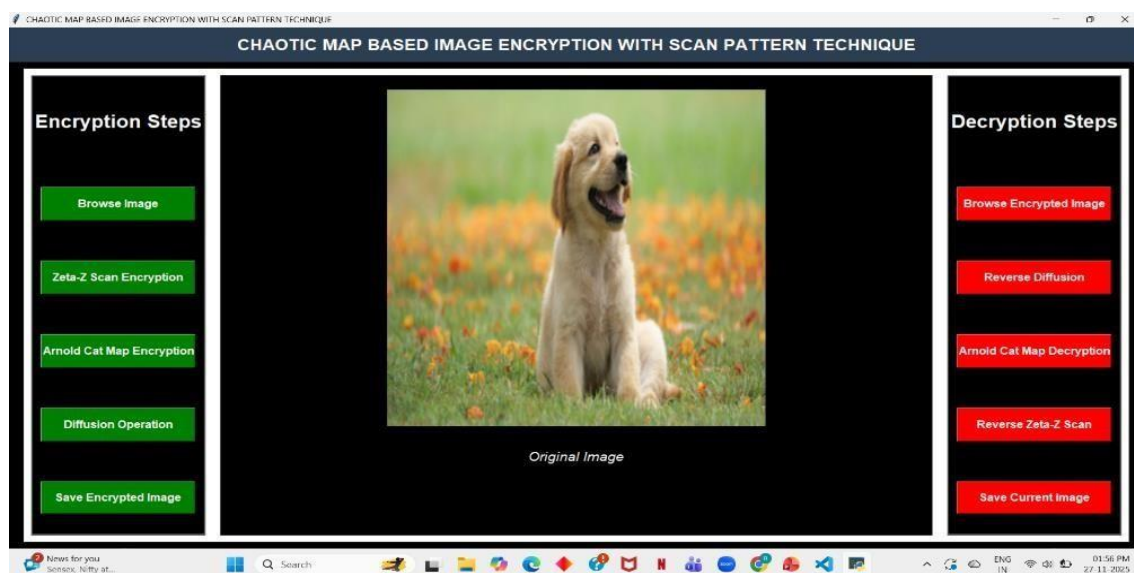


Fig 4.9 : Use of Browse Button

The scan pattern encrypted image button is used to perform the zeta z scan operation on input image as shown in the Fig 4.10.

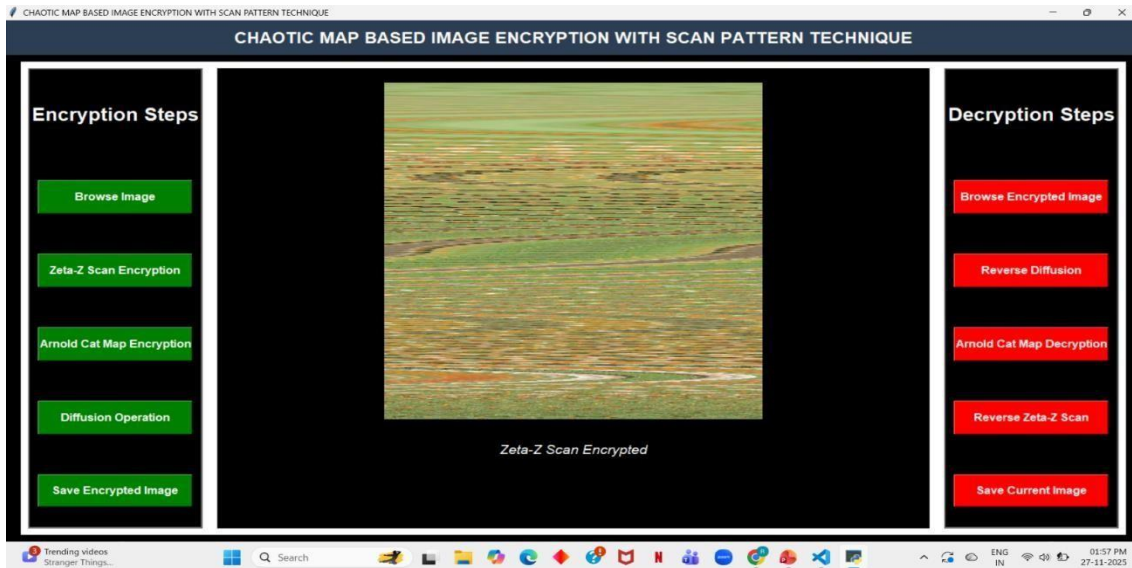


Fig 4.10 : Use of Scan pattern encrypted image button

The scan image is confused using Arnold's cat map by using Arnold's cat map encryption button as shown in the Fig 4.11.

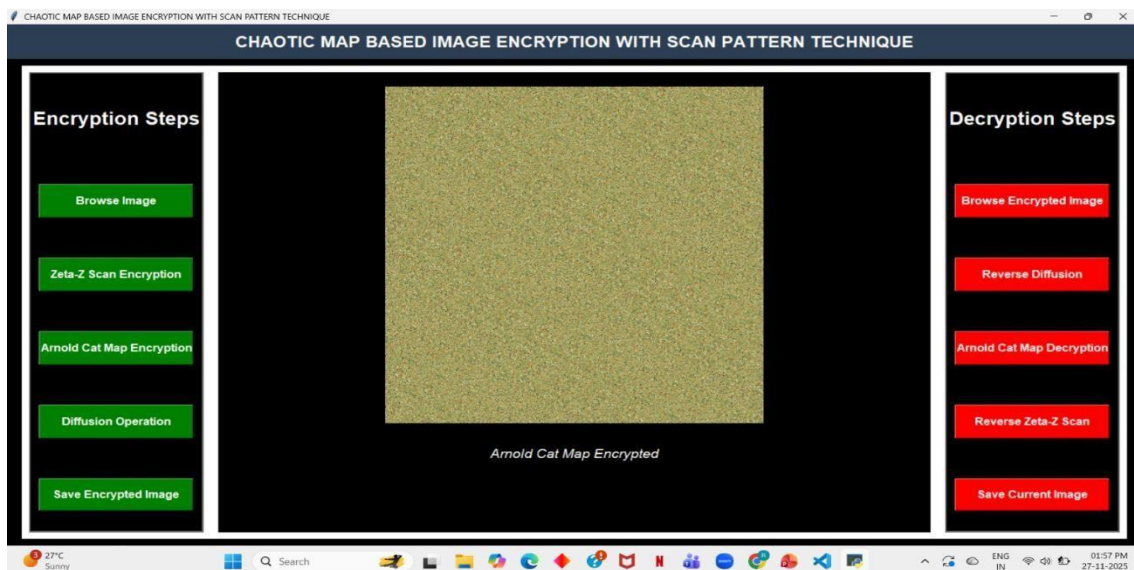


Fig 4.11 : Use of Arnold's cat map encryption button

The diffusion operation applied on confused image to obtain cipher image using diffusion operation button as shown in Fig 4.12.

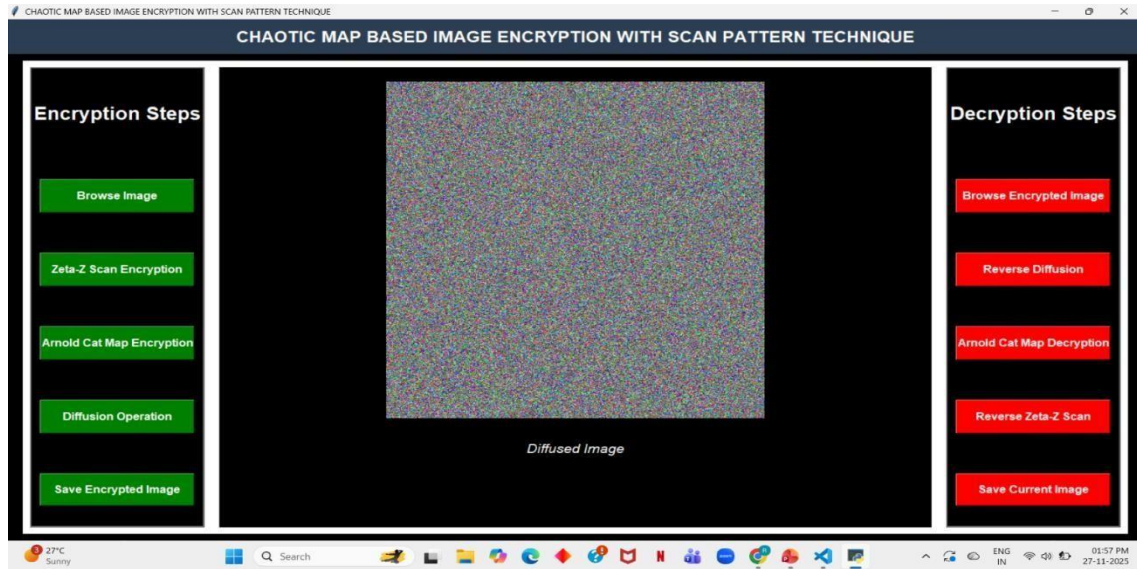


Fig 4.12 : Use of Diffusion operation button

Then obtained cipher image is used as input image for decryption process and performing reverse of diffusion operation using reverse diffusion operation button as shown in the Fig 4.13.

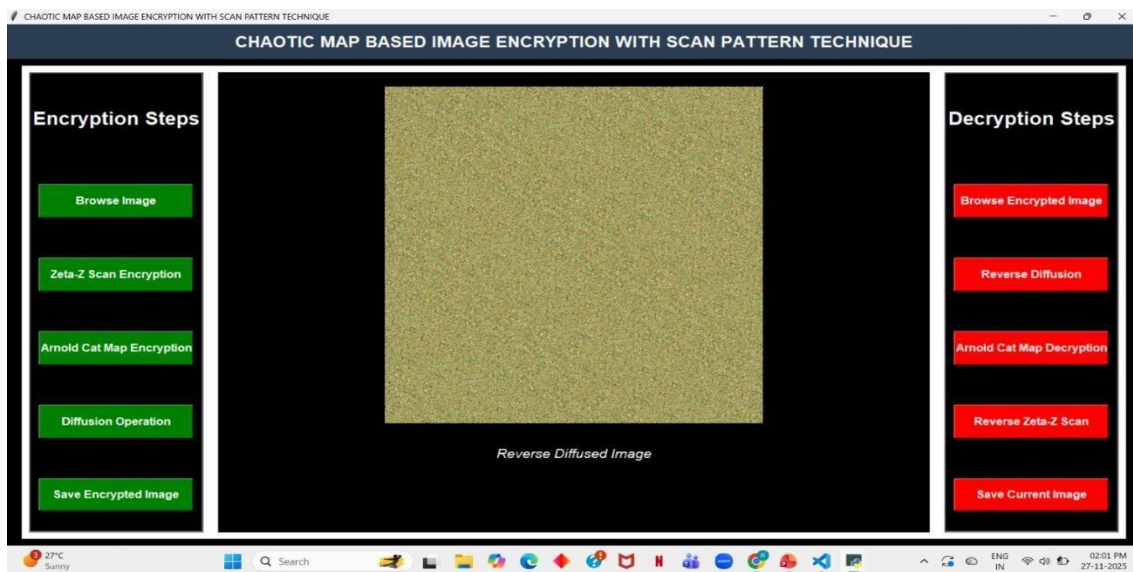


Fig 4.13 : Use of Reverse Diffusion operation button

Then performing reverse of Arnold's cat map to obtain the scan image by using Arnold's cat map decryption button as shown in the figure 4.14.

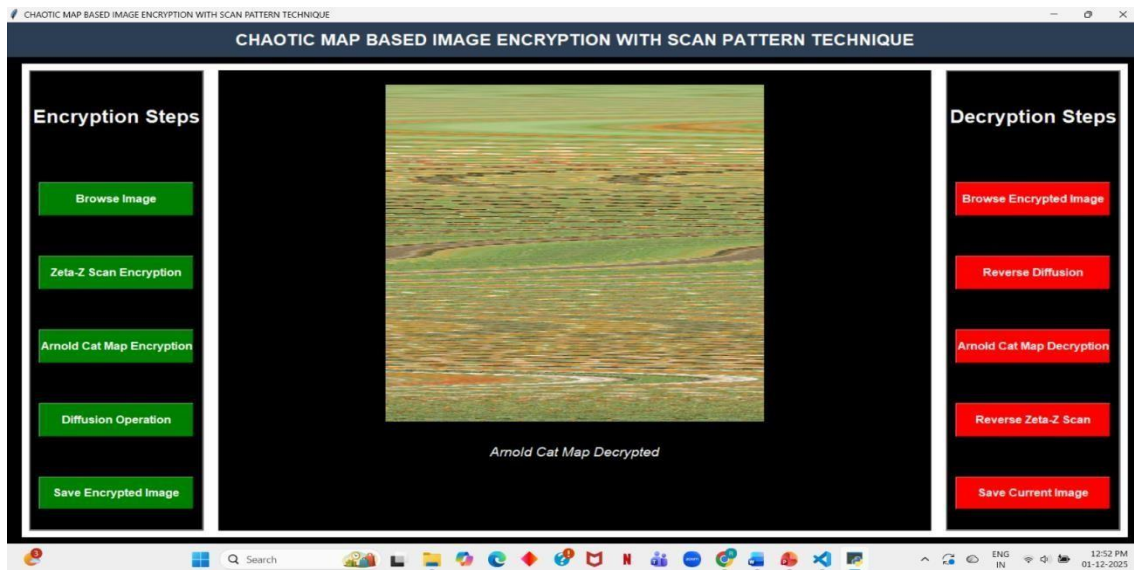


Fig 4.14 : Use of Arnold's cat map decryption button

Performing reverse of scan operation to obtain original image by using reverse scan operation button as shown in Fig 4.15.

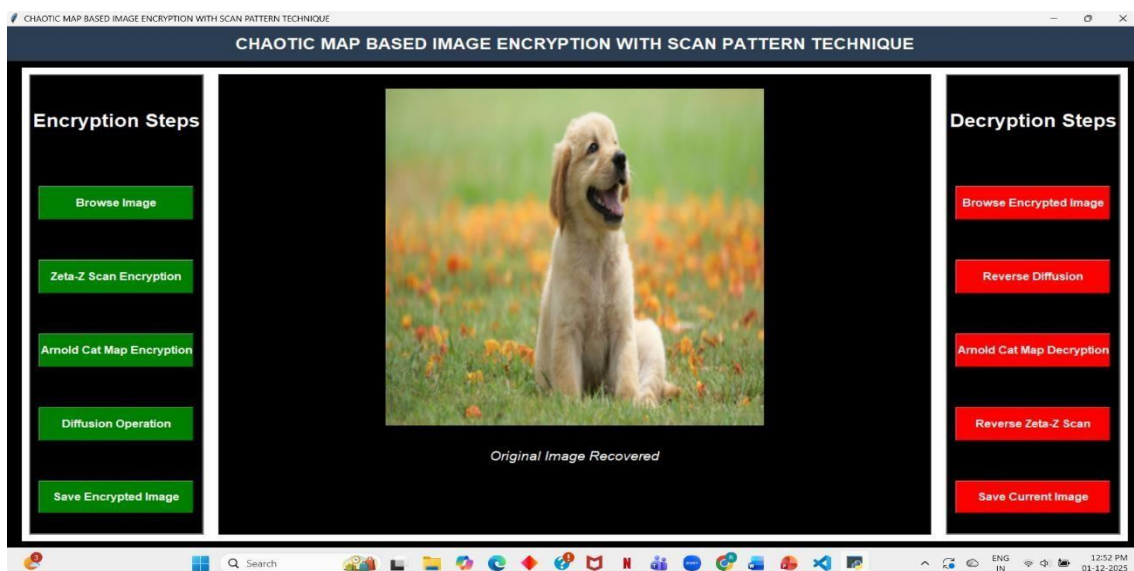


Fig 4.15 : Use of Reverse of scan pattern button

4.1 RESULT ANALYSIS

An encryption technique is only useful if it can sustain attacks from intruders, interceptors, or any other form of enemies. An encryption algorithm is known by its strength. Therefore, it becomes necessary to test any encryption processing before it can be applied in the live environment. Otherwise, it would be a wastage of both time and useful resources. Five types of tests are used to measure the strength of the proposed algorithm. These tests are:

4.2.1 Correlation coefficient

Correlation analysis is a method for finding the relation between the original and the secure image here. It is used to determine how the change in one variable affects the other in terms of degree and direction. By quantifying the intensity and direction of the relationship between variables, correlation analysis offers important insights, such as how interdependent they are. Correlation analysis is a valuable statistical approach used to investigate and quantify links between variables in various fields. The correlation coefficients are displayed in Table 4.1 represents the correlation analysis of the original and secure images.

SPQ - Correlation coefficient.

$$SPQ = \text{cov}(P, Q) / \sqrt{D(P)}\sqrt{D(Q)}$$



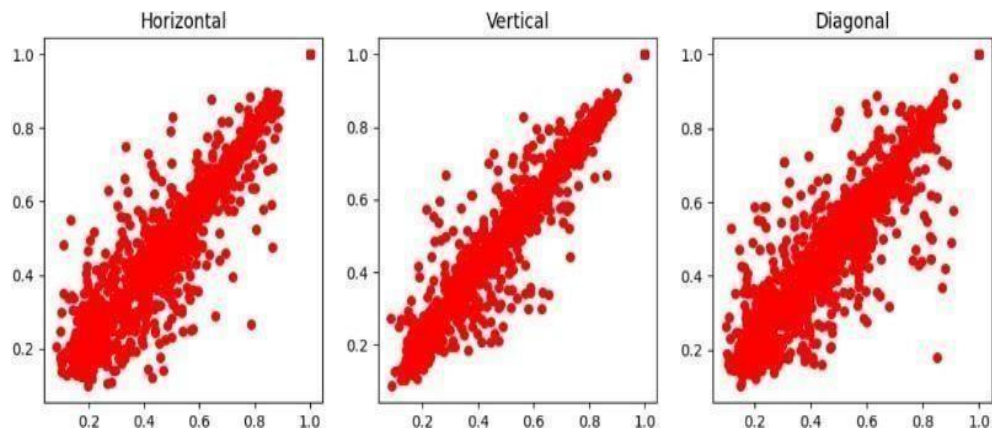


Fig 4.16 : Correlation coefficient of original image

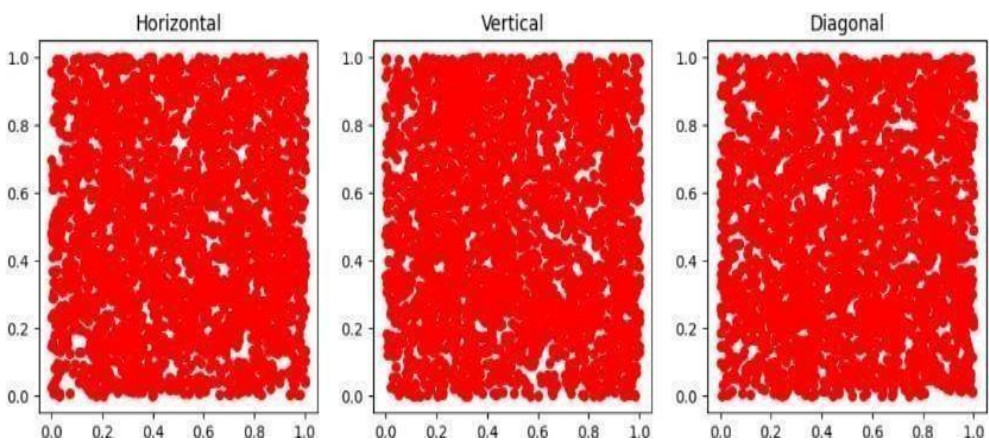


Fig 4.17 : Correlation coefficient of cipher image

Correlation coefficient for different images is calculated using equation 4 and listed in Table 4.1. From the below list we can specify that correlation coefficient for the original image is near to 1 and for the cipher image it is nearer to the 0.1. So, it shows that the proposed scheme is highly resistant to the attacks.

Table 4.1 : Correlation coefficient of original image and cipher image

Image	Horizontal		Vertical		Diagonal	
	Original image	Cipher image	Original image	Cipher image	Original image	Cipher image
Lena	0.9952	-0.02743	0.9976	0.01822	0.9919	0.00923
Cameraman	0.9952	0.03173	0.9976	-0.00059	0.9921	-0.01297
Baboon	0.9950	-0.00134	0.9948	-0.00821	0.9898	0.00532
Dog	0.9983	0.03967	0.9985	-0.02104	0.9968	0.03960

4.2.2 The Information Entropy Analysis

The information entropy is one of the standard measures that used to measure the intensity of a symmetric cryptosystem. The information entropy $H(m)$ of source message m is defined by (5). The perfect value of information entropy for an 8-bit message is 8, which means that the message source m can be counted as random.

$$E = \sum_{i=0}^n -1 [P(i) * \log_2 \frac{1}{P(i)}] \quad (5)$$

where $P(i)$ represents the probability of occurrence of the symbol. The entropies of plain and encrypted images are shown in Table 4.2. The results in this table show that the entropies of the encrypted image which is nearer to the 8 generated by the proposed algorithm.

Table 4.2 : Information Entropy

Entropy	Original image	Cipher image
Lena	7.3566823	7.4329357
Cameraman	7.0487328	7.2826304
Baboon	7.1449323	7.3466725
Dog	7.3070145	7.4643126

4.2.3 NPCR Analysis (Number of Pixel Change Rate)

The meaning of NPCR the change rate of pixels in encrypted image when the value of one pixel in the original image is changed. It measures the percentage of different pixel numbers between the cipher images while their corresponding original images have only one different pixel. Let the cipher images be P1 and P2. Define a bipolar array Q, with the same size as image P1 or P2. If $P1(i, j) \neq P2(i, j)$ then $Q(i, j)=1$; otherwise, $Q(i, j)=0$. The NPCR is defined as

$$NPCR = \frac{\sum_{i=1}^H \sum_{j=1}^W Q(i,j)}{W*H} * 100\% \quad (6)$$

The NPCR calculate the percentage of different pixel numbers between two images. This analysis helps in determining the vulnerability of the system towards differential attacks. The analysis report for encrypted image is given in the Table 4.3. In which the NPCR value is nearer to the 99 percent so it shows that proposed method is good for securing image from the attacks.

Table 4.3 : NPCR Analysis

Image	NPCR
Lena	99.2%
Cameraman	99.5%
Baboon	98.9%
Dog	99.1%

4.2.4 UACI Analysis (Unified Average Changing Intensity)

UACI calculate the average intensity difference between plain text image and encrypted image. The mathematical formulation is

$$UACI = \frac{1}{W*H} \left[\sum_{i=1}^H \sum_{j=1}^W \frac{|P1(i,j)-P2(i,j)|}{255} \right] * 100\% \quad (7)$$

UACI calculates the average intensity of differences between two images. The analysis reported for encrypted image is given in the Table 4.4. The results in this table shows that the UACI performance of the proposed method relatively nearer to the value of 33 percent it is good method for securing images from the attacks.

Table 4.4 : UACI Analysis

Image	UACI
Lena	20.45%
Cameraman	24.7%
Baboon	18.1%
Dog	24.9%

4.2.5 SSIM Analysis

The SSIM between the input (original) image and the decrypted image should be 1 for a good encryption technique. This makes SSIM a better indicator of perceived image quality. Higher SSIM values indicate greater similarity between images, with a maximum value of 1 indicating identical images. The proposed algorithm SSIM values as 1 listed in Table 4.5. This can be useful in tasks like image compression, where you want to minimize loss of quality, or in image restoration, where you want to assess the effectiveness of the restoration process

Table 4.5 : SSIM Analysis

Image	SSIM
Lena	1.0
Cameraman	1.0
Baboon	1.0
Dog	1.0

4.2.6 Mean Square Error and Peak Signal-To-Noise Ratio (MSE & PSNR)

The mean-square error (MSE) and the peak signal-to-noise ratio (PSNR) are used to compare image compression quality. The MSE represents the cumulative squared error between the compressed and the original image.

Whereas, PSNR represents a measure of the peak error. The lower the value of MSE, the lower the error.

The equation to calculate MSE is as given below

$$MSE = \frac{\sum_{M,N} [I_1(m,n) - I_2(m,n)]^2}{M \times N} \quad (8)$$

Where, M and N are the number of rows and columns in the input images.

The equation to calculate PSNR is as given below

$$PSNR = 10 \log_{10} \left(\frac{R^2}{MSE} \right) \quad (9)$$

Where, R is the maximum fluctuation in the input image data type. Table 4.6 lists the values for MSE and PSNR. The Table 4.6 shows that the value of MSE is greater and PSNR value is lower so it is good method securing image from the differential attacks.

Table 4.6 : MSE and PSNR Analysis

Image	MSE	PSNR
Lena	21749.289	4.7563
Cameraman	28791.253	3.5382
Baboon	18349.041	5.4946
Dog	28679.908	3.5550

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

The conclusion and future scope of the project is explained below in this section.

5.1 CONCLUSION

The proposed algorithm consists of four stages that are scan operation, confusion, generation of encryption key sequence and diffusion. The original image undergoes a series of intricate operations to ensure robust security. Beginning with the selection of the image, its dimensions are utilized in a Zeta Z scan pattern, initiating a thorough examination. This scan output then encounters the chaotic transformations of Arnold's cat map through the Confusion operation, adding complexity to the image structure. Random key sequences, crucial for encryption, are generated via the Logistics map, further enhancing the cryptographic strength. Subsequently, the Diffusion operation shuffles pixels using the Spiral-In scan pattern, adding another layer of disorder. The final step involves an XOR operation between the Spiral-In scan pattern's pixel values and the generated random key sequences, creating the Cipher image. Through these meticulously orchestrated steps, the encryption process ensures the original image's transformation into a highly secure and cryptographically protected form, safeguarding it against unauthorized access and maintaining data integrity.

5.2 FUTURE SCOPE

The present scan operation and chaotic system has good prospects for applications in information encryption and a secure communication channel is designed for the encryption of colour images. The future scope includes extending to the n number of layers of encryption and decryption process for colour images for security purpose.

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PUBLICATION DETAILS AND CERTIFICATE



International Journal of Scientific Research in Engineering and Management (IJSREM)
Volume: 09 Issue: 10 | Oct - 2025 SJIF Rating: 8.586 ISSN: 2582-3930

Chaotic Map Based Image Encryption with Scan Pattern Technique

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Abstract— This work proposes an image encryption method combining chaotic maps and a dynamic scan pattern for enhanced security. Chaotic maps, with their pseudorandom behavior, generate cryptographic keys that add unpredictability to the encryption process. These keys dynamically influence the encryption algorithm, increasing security. Additionally, a unique scan pattern is used to shuffle pixel values, improving the diffusion of information. The method starts with the chaotic map to generate keys and then applies the scan pattern to redistribute pixels, further strengthening the encryption.

Algorithm, " uses SCAN patterns with XTEA encryption for efficient image security, though it can cause distortion and XTEA has known vulnerabilities. These methods provide robust image encryption with varying trade-offs in complexity and performance.

The proposed image encryption techniques use chaotic systems to enhance security. "An Image Encryption Technique using Logistic Map and Z-order Curve" combines the chaotic Logistic map and Z-order curve to generate a key stream based on the image, offering strong encryption but potential vulnerability to sophisticated attacks. "An Image Cryptography using Henon Map and Arnold Cat Map" uses Henon's chaotic system and the Arnold Cat Map for pixel shuffling, improving security with a computational cost. "Chaos-Based Confusion and Diffusion of Image Pixels Using Dynamic Substitution" applies Henon and Ikeda maps with dynamic substitution for enhanced pixel confusion and diffusion, ensuring high security but with increased computational complexity. These methods all leverage chaos theory for secure image encryption but face challenges with computational overhead and potential vulnerabilities.

The studies on image encryption using chaotic maps explore methods to secure image data, particularly in sectors like telemedicine and digital communication. One approach uses Henon chaotic maps, offering strong security but facing implementation and computational complexity challenges. Another combines Logistic, Tent, and Sine maps for better encryption, though efficiency is reduced. A system for tele-dermatology employs Arnold's cat map and Henon map for confusion and diffusion, providing strong security but with high computational demands. A hybrid encryption method using these maps improves security but slows processing speed. Additionally, a survey compares various chaotic maps, highlighting their strengths and weaknesses, though some methods struggle with speed. These techniques enhance data security but present efficiency and real-time application challenges.

I. INTRODUCTION

With the rapid growth of data communication, securing sensitive information is crucial. Cryptographic techniques are needed to protect data before transmission, especially for images, which are larger and more complex than text. Traditional encryption methods like DES, AES, and RSA are inefficient for image encryption due to size and redundancy.

This work introduces a chaotic map-based encryption approach for images, leveraging the nonlinearity and unpredictability of chaos theory. Chaotic maps generate pseudorandom streams or control parameters, making them suitable for both stream and block ciphers. Chaos-based encryption offers fast, secure implementation with high sensitivity and efficiency.

II. LITERATURE SURVEY

The three image encryption techniques enhance security through cryptography and steganography. The first method, "Image Encryption using Scan Patterns and Hill Cipher," uses SCAN patterns and the Hill cipher for multi-layered encryption, improving security but adding complexity and overhead. The second, "Triple-Layer Image Security Using a Zigzag Embedding Pattern," combines AES-128, chaotic encryption, and zigzag-based steganography, offering strong security and high image quality but being computationally expensive. The third method, "Image Encryption and Decryption Using Scan Pattern and XTEA Encryption









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Department of Computer Science and Engineering

Project- Sustainable Development Goals (SDG) Mapping

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Project Title	Chaotic Map Based Image Encryption with Scan Pattern Technique
SDG 1: No Poverty	☐
SDG 2: Zero Hunger	☐
SDG 3: Good Health and Well-Being	☐
SDG 4: Quality Education	☐
SDG 5: Gender Equality	☐
SDG 6: Clean Water and Sanitation	☐
SDG 7: Affordable and Clean Energy	☐
SDG 8: Decent Work and Economic Growth	☐
SDG 9: Industry, Innovation, and Infrastructure	☒
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